DISCRETE STRUCTURES (CSC 203)

A

LECTURE NOTE PREPARED

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FOR

DEGREE PROGRAMME

Propositional Logic

A field of mathematics known as propositional logic examines the logical connections between propositions—also known as statements, sentences, or assertions—when viewed as a whole and joined by logical connectives.

Example:

1. The sun rises in the East and sets in the West.
2. 1 + 1 = 2
3. 'b' is a vowel.

Above 3 sentences are propositions, where the first two are Valid (True) and the third one is Invalid (False).

All of the above sentences are propositions, where the first two are Valid (True) and the third one is Invalid (False). Some sentences that do not have a truth value or may have more than one truth value are not propositions.

Example,

- What time is it?
- Go out and Play
- x + 1 = 2

Third sentence may or may not be true, and the first two are not propositions since they lack truth value. Propositional variables, such as p, q, r, and s, are used to represent propositions. Propositional calculus, also known as propositional logic, is the area of logic that focusses on propositions. It also addresses the creation of new hypotheses from preexisting ones. Compound propositions are propositions composed of one or more propositions joined by logical connectives (also known as operators).

Types of Propositions

Propositions are statements that can be assessed as true or false in propositional logic. They serve as the foundation for more intricate logical arguments. The two primary categories of propositions are broken down as follows:

- Atomic Propositions
- Compound Propositions

Atomic Propositions in Propositional Logic

The fundamental unit of propositional logic is an atomic proposition, sometimes referred to as a simple proposition.

- It is a declarative sentence that, according to propositional logic, cannot be divided into smaller statements but is either true (T) or false (F).
- Logical connectives such as AND (∧), OR (∨), NOT (¬), etc., are absent from atomic propositions.

Examples of Atomic Propositions

1. **P:** "It is raining."
2. **Q:** "The sun is shining."

3. **R:** "2 + 2 = 4."

Each of these is a complete, indivisible statement — it's either true or false.

Non-Atomic (Compound) Propositions

When we combine atomic propositions with logical connectives, we get **compound propositions**.

For example:

- **P ∧ Q** → "It is raining AND the sun is shining."
- **¬P** → "It is NOT raining."
- **P → Q** → "If it is raining, then the sun is shining."

Here, **P** and **Q** are atomic propositions, while **P ∧ Q**, **¬P**, and **P → Q** are compound propositions.

Truth Table of Propositional Logic

We take into account every possible combination of propositions that are joined together by logical connectives to form the given compound proposition because we need to know the truth value of a proposition in every scenario. A truth table is a tabular compilation of every scenario that could possibly occur. Most Frequently Used Logical Connectives:

1. Negation

If p is a proposition, then the negation of p is denoted by \neg p, which when translated to simple English means- "It is not the case that **p**" or simply "not **p**". The truth value of **-p** is the opposite of the truth value of **p**. The truth table of **-p** is:

p	**¬p**
T	F
F	T

Example, Negation of "It is raining today", is "It is not the case that is raining today" or simply "It is not raining today".

2. Conjunction

For any two propositions p and q, their conjunction is denoted by p\wedge q, which means "p and q ". The conjunction p\wedge q is True when both p and q are True, otherwise False. The truth table of p\wedge q is:

p	**q**	**p ∧ q**
T	T	T
T	F	F
F	T	F
F	F	F

Example, Conjunction of the propositions p - "Today is Friday" and q - "It is raining today", p\wedge q is "Today is Friday and it is raining today". This proposition is true only on rainy Fridays and is false on any other rainy day or on Fridays when it does not rain.

3. Disjunction

For any two propositions p and q, their disjunction is denoted by p\vee q, which means "p or q ". The disjunction p\vee q is True when either p or q is True, otherwise False. The truth table of p\vee q is:

p	**q**	**p ∨ q**
T	T	T
T	F	T
F	T	T
F	F	F

Example, Disjunction of the propositions p - "Today is Friday" and q - "It is raining today", p\vee q is "Today is Friday or it is raining today". This proposition is true on any day that is a Friday or a rainy day (including rainy Fridays) and is false on any day other than Friday when it also does not rain.

4. Exclusive Or

For any two propositions p and q , their exclusive or is denoted by p\oplus q , which means "either p or q but not both". The exclusive or p\oplus q is True when either p or q is True, and False when both are true or both are false. The truth table of p\oplus q is:

p	**q**	**p ⊕ q**
T	T	F
T	F	T
F	T	T
F	F	F

Example, Exclusive or of the propositions p - "Today is Friday" and q - "It is raining today", p\oplus q is "Either today is Friday or it is raining today, but not both". This proposition is true on any day that is a Friday or a rainy day (not including rainy Fridays) and is false on any day other than Friday when it does not rain or rainy Fridays.

5. Implication

For any two propositions p and q , the statement "if p then q " is called an implication and it is denoted by p\rightarrow q . In the implication p\rightarrow q , p is called the **hypothesis** or **antecedent** or **premise** and q is called the **conclusion** or **consequence**. The implication is p\rightarrow q is also called a **conditional statement**. The implication is false when p is true and q is false otherwise it is true. The truth table of p\rightarrow q is:

p	**q**	**p → q**
T	T	T
T	F	F
F	T	T
F	F	T

One might wonder that why is p\rightarrow q true when p is false. This is because the implication guarantees that when p and q are true then the implication is true. But the implication does not guarantee anything when the premise p is false. There is no way of knowing whether or not the implication is false since p did not happen. This situation is similar to the "Innocent until proven Guilty" stance, which means that the implication p\rightarrow q is considered true until proven false. Since we cannot call the implication p\rightarrow q false when p is false, our only alternative is to call it true.

This follows from the **Explosion Principle** which says: "A False statement implies anything" Conditional statements play a very important role in mathematical reasoning, thus a variety of terminology is used to express p\rightarrow q, some of which are listed below.

"If p, then "q"p is sufficient for q""q when p""a necessary condition for p is q""p only if q""q unless ≠p""q follows from p"

Example, "If it is Friday then it is raining today" is a proposition which is of the form p\rightarrow q. The above proposition is true if it is not Friday (premise is false) or if it is Friday and it is raining, and it is false when it is Friday but it is not raining.

6. Biconditional or Double Implication

For any two propositions p and q, the statement "p if and only if(iff) q " is called a biconditional and it is denoted by p\leftrightarrow q. The statement p\leftrightarrow q is also called a **bi-implication**. p\leftrightarrow q has the same truth value as (p\rightarrow q) \wedge (q\rightarrow p). The implication is true when p and q have same truth values, and is false otherwise. The truth table of p\leftrightarrow q is:

p	**q**	**p ↔ q**
T	T	T
T	F	F
F	T	F
F	F	T

Some other common ways of expressing p\leftrightarrow q are:
"p is necessary and sufficient for q""if p then q, and conversely""p if q"
Example, "It is raining today if and only if it is Friday today." is a proposition which is of the form p\leftrightarrow q . The above proposition is true if it is not Friday and it is not raining or if it is Friday and it is raining, and it is false when it is not Friday or it is not raining.

Tautologies
A Tautology is a formula which is always true for every value of its propositional variables.

Example − Prove $[(A \rightarrow B) \wedge A] \rightarrow B$ is a tautology
The truth table is as follows –

A	**B**	**A → B**	**(A → B) ∧ A**	**[(A → B) ∧ A] → B**
True	True	True	True	True
True	False	False	False	True
False	True	True	False	True
False	False	True	False	True

As we can see every value of $[(A \rightarrow B) \wedge A] \rightarrow B$ is "True", it is a tautology.

Contradictions
A Contradiction is a formula which is always false for every value of its propositional variables.
Example − Prove $(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$ is a contradiction
The truth table is as follows –

A	**B**	**A ∨ B**	**¬ A**	**¬ B**	**(¬ A) ∧ (¬ B)**	**(A ∨ B) ∧ [(¬ A) ∧ (¬ B)]**
True	True	True	False	False	False	False
True	False	True	False	True	False	False
False	True	True	True	False	False	False
False	False	False	True	True	True	False

As we can see every value of $(A \vee B) \wedge [(\neg A) \wedge (\neg B)]$ is False, it is a contradiction.

Contingency
A Contingency is a formula which has both some true and some false values for every value of its propositional variables.
Example − Prove $(A \vee B) \wedge (\neg A)$ a contingency
The truth table is as follows –

A	**B**	**A ∨ B**	**¬ A**	**(A ∨ B) ∧ (¬ A)**
True	True	True	False	False
True	False	True	False	False
False	True	True	True	True
False	False	False	True	False

As we can see every value of $(A \vee B) \wedge (\neg A)$ has both True and False, it is a contingency.

Propositional Equivalences

Two statements X and Y are logically equivalent if any of the following two conditions hold −

- The truth tables of each statement have the same truth values.
- The bi-conditional statement $X \Leftrightarrow Y$
- is a tautology.

Example − Prove $\neg(A \vee B)$ *and* $[(\neg A) \wedge (\neg B)]$ are equivalent

Testing by 1st method (Matching truth table)

A	B	A ∨ B	¬ (A ∨ B)	¬ A	¬ B	[(¬ A) ∧ (¬ B)]
True	True	True	False	False	False	False
True	False	True	False	False	True	False
False	True	True	False	True	False	False
False	False	False	True	True	True	True

Here, we can see the truth values of $\neg(A \vee B)$ *and* $[(\neg A) \wedge (\neg B)]$ are same, hence the statements are equivalent.

Testing by 2nd method (Bi-conditionality)

A	B	¬ (A ∨ B)	[(¬ A) ∧ (¬ B)]	[¬ (A ∨ B)] ⇔ [(¬ A) ∧ (¬ B)]
True	True	False	False	True
True	False	False	False	True
False	True	False	False	True
False	False	True	True	True

As $[\neg(A \vee B)] \Leftrightarrow [(\neg A) \wedge (\neg B)]$ is a tautology, the statements are equivalent.

Inverse, Converse, and Contra-positive

Implication / if-then (→)

is also called a conditional statement. It has two parts −

- Hypothesis, p
- Conclusion, q

As mentioned earlier, it is denoted as $p \rightarrow q$

.

Example of Conditional Statement − If you do your homework, you will not be punished. Here, "you do your homework" is the hypothesis, p, and "you will not be punished" is the conclusion, q.

Inverse − An inverse of the conditional statement is the negation of both the hypothesis and the conclusion. If the statement is If p, then q, the inverse will be If not p, then not q. Thus, the inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$

.

Example − The inverse of If you do your homework, you will not be punished is If you do not do your homework, you will be punished.

Converse − The converse of the conditional statement is computed by interchanging the hypothesis and the conclusion. If the statement is If p, then q, the converse will be If q, then p. The converse of $p \rightarrow q$
is $q \rightarrow p$
.
Example − The converse of "If you do your homework, you will not be punished" is "If you will not be punished, you do your homework.

Contra-positive − The contra-positive of the conditional is computed by interchanging the hypothesis and the conclusion of the inverse statement. If the statement is If p, then q, the contra-positive will be If not q, then not p. The contra-positive of $p \rightarrow q$
is $\neg q \rightarrow \neg p$
.
Example − The Contra-positive of " If you do your homework, you will not be punished is "If you are punished, you did not do your homework.

Duality Principle
Duality principle states that for any true statement, the dual statement obtained by interchanging unions into intersections (and vice versa) and interchanging Universal set into Null set (and vice versa) is also true. If dual of any statement is the statement itself, it is said **self-dual** statement.

Example − The dual of $(A \cap B) \cup C$
is $(A \cup B) \cap C$

Normal Forms
We can convert any proposition in two normal forms −

- Conjunctive normal form
- Disjunctive normal form

Conjunctive Normal Form
A compound statement is in conjunctive normal form if it is obtained by operating AND among variables (negation of variables included) connected with ORs. In terms of set operations, it is a compound statement obtained by Intersection among variables connected with Unions.
Examples

- $(A \vee B) \wedge (A \vee C) \wedge (B \vee C \vee D)$
- $(P \cup Q) \cap (Q \cup R)$

A compound statement is in disjunctive normal form if it is obtained by operating OR among variables (negation of variables included) connected with ANDs. In terms of set operations, it is a compound statement obtained by Union among variables connected with Intersections.
Examples

- $(A \wedge B) \vee (A \wedge C) \vee (B \wedge C \wedge D)$
- $(P \cap Q) \cup (Q \cap R)$

Logical Equivalence in Discrete Mathematics

Logical equivalence is a fundamental concept in propositional logic. It is used in analyzing and transforming logical statements into more manageable forms. Logical equivalence is the relationship between two statements that have the same truth value in every possible scenario. In simpler terms, we can say two statements are logically equivalent if they are both either True or False. It is like saying two people always agreeing on every topic.

Mathematically, we can express this relationship between two statements **P** and **Q** by writing **P ≡ Q**, which means **P** and **Q** are logically equivalent.

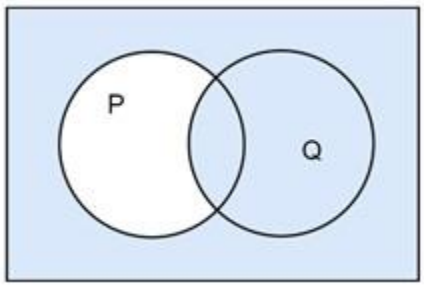
Understanding Logical Equivalence Using Truth Tables

Like logics, a truth table is a tool to determine whether two statements are logically equivalent. We list out all possible truth values for each statement and then compare the results. If the truth values match in every case, the statements are logically equivalent.

Example of Logical Equivalence of P → Q and ¬P ∨ Q

Let us see an example of logical equivalence between two statements, **P → Q** (if P then Q) and **¬P ∨ Q** (not P or Q). To prove these two statements are logically equivalent, we construct the truth table –

P	**Q**	**P → Q**	**¬P ∨ Q**
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T



As we can see, the final columns of both statements are the same, meaning **P → Q** and **¬P ∨ Q** are **logically equivalent**.

De Morgan's Laws

We have learnt the concept of De Morgan's Laws in Set Theory and Boolean algebra. They are a set of logical equivalences in discrete mathematics. These laws show us a way to distribute negations across conjunctions (ANDs) and disjunctions (ORs). They are useful in simplifying complex logical expressions.

De Morgan's Laws are −

- ¬ (P ∧ Q) ≡ ¬P ∨ ¬Q
- ¬ (P ∨ Q) ≡ ¬P ∧ ¬Q

Example of Logical Equivalence Using De Morgan's Law

Let us see how to apply De Morgan's law to check if the statements **¬(P ∨ Q)** and **¬P ∧ ¬Q** are logically equivalent.

P	**Q**	**¬ (P ∨ Q)**	**¬P ∧ ¬Q**
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

As the truth values are the same, De Morgan's law holds, confirming that **¬(P ∨ Q) ≡ ¬P ∧ ¬Q**

Negation of Implications

Another useful but trickier thing is negation of implications. The negation of an implication **P → Q** is not another implication. It is a conjunction of the hypothesis being true and the conclusion being false. In other words, **¬ (P → Q) ≡ P ∧ ¬Q**

Example of Negating an Implication

Let us see the prove that **¬ (P → Q) ≡ P ∧ ¬Q** without using a truth table by breaking it down step-by-step −

- Start with ¬ (P → Q).
- Recall that P → Q is equivalent to ¬P ∨ Q (We also proved this in the first example)
- Now apply negation: ¬(¬P ∨ Q)
- Using De Morgan's Law, we get ¬¬P ∧ ¬Q
- Finally, simplify the double negation to P ∧ ¬Q

By using logical transformations instead of truth tables, we showed that **¬(P → Q) ≡ P ∧ ¬Q**

More Complex Examples of Logical Equivalence

We have understood the basics examples. Let us see some of the complex examples where we have much things to do. Let us determine whether it is logically equivalent to another statement using a truth table.

Example: Are (P ∨ Q) → R and (P → R) ∨ (Q → R) Logically Equivalent?

Constructing the truth table –

P	**Q**	**R**	**(P ∨ Q) → R**	**(P → R) ∨ (Q → R)**
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	F	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

In the fourth and sixth rows, the truth values differ, meaning that the two statements are not logically equivalent. However, in certain scenarios, $(P \vee Q) \rightarrow R$ implies $(P \rightarrow R) \vee (Q \rightarrow R)$, but not the reverse.

Predicates and Quantifiers

Predicates and Quantifiers are fundamental concepts in mathematical logic, essential for expressing statements and reasoning about the properties of objects within a domain. These concepts are widely used in computer science, engineering, and mathematics to formulate precise and logical statements.

Predicates

A predicate is a statement that contains variables and becomes a proposition when specific values are substituted for those variables. Predicates express properties or relations among objects. Example:

P(x) = "x is an even number"

- When x = 2, P (2) is True.
- When x = 3, P (3) is False.

A predicate is a logical expression that contains variables and becomes a statement when specific values are substituted for those variables. So, predicates describe properties or conditions that elements of a set can satisfy.

Example of a Predicate

Consider we have a predicate **P(x)** that states, "**x is a prime number**".

- If we replace **x** with 2, the predicate **P (2)** becomes the statement, "**2 is a prime number**" which is True.
- Similarly, if we substitute **x = 4**, the predicate P (4) becomes "**4 is a prime number**" which is False.

In both the cases, by substituting specific values for **x**, the predicate becomes a definite statement that can be either True or False.

Difference between a Predicate and a Statement

Predicates and Statements are quite similar and that's why they become confusing. A **statement** is a sentence that is either True or False. However, a **predicate** has a variable and is not a statement until the variable is replaced with a specific value. For **example**:

- **Predicate** − "n is a prime number."
- **Statement** − "7 is a prime number."

Without assigning a value to the variable **n**, the predicate is incomplete and does not hold a truth value. Only when we provide a value for **n**, like in the case of "7 is a prime number," does the predicate become a true or false statement.

Quantifiers in Discrete Mathematics

The next part is the quantifiers. These are symbols or words that define the extent to which a predicate is true for a set of elements.
In mathematics, we use two main types of quantifiers:

- **Universal quantifier** (denoted by ∀), meaning "for all."
- **Existential quantifier** (denoted by ∃), meaning "there exists."

Universal Quantifier (∀)
From the name itself we can understand this. The universal quantifier is used when we want to declare that a predicate is true for every possible value of a variable. It is read as "for all" or "for every."
Let us understand with **examples**. Consider the following statement −
$\forall x\ (x \geq 0)$
It means, "For all values of **x**, **x** is greater than or equal to zero." In the domain of natural numbers (0, 1, 2, ...), this statement is True.

Existential Quantifier (∃)
Another type of quantifier is the existential quantifier. It is used when we want to say that there is at least one value of a variable for which a predicate is true. It is read as "there exists" or "there is."
For example, consider the following statement −
$\exists x\ (x < 0)$
This means "There exists an x such that x is less than zero." In the domain of natural numbers, this statement is false because no natural number is less than zero. However, in the domain of real numbers, this statement is true.

Combining the Quantifiers
We can combine these two quantifiers for better use cases. Let us look at how this works by checking the following examples –

Example of Universal and Existential Quantifiers
Consider the statement −
$\forall x\ \exists y\ (y < x)$
This reads, "For every x, there exists a y such that y is less than x."
If we are working within the domain of natural numbers (0, 1, 2, ...), this statement is **false** because for x = 0, there is no natural number y such that y < 0. However, if we change the domain to real numbers, this statement becomes **true**, since for any real number x, there exists a real number y that is smaller.

Example of Nested Quantifiers
Now, let us look at a more complex statement with nested quantifiers −
$\exists x\ \forall y\ (y \geq x)$
This reads, "There exists an **x** such that for all **y**, **y** is greater than or equal to **x**."
In the domain of natural numbers, this statement is **true** because the number **x = 0** satisfies this condition. No matter which number **y** we choose, it will always be greater than or equal to 0.

Negation of Quantifiers

When we negate quantified statements, the type of quantifier changes. The negation of a universal quantifier becomes an existential quantifier, and the negation of an existential quantifier becomes a universal quantifier.

Following are the **rules of negation** −

- **¬∀xP(x)** is equivalent to **∃x¬P(x)**, meaning "It is not true that for all **x**, **P(x)** is true" is the same as "There exists an **x** such that **P(x)** is false."
- **¬∃xP(x)** is equivalent to **∀x¬P(x)**, meaning "There does not exist an **x** such that **P(x)** is true" is the same as "For all **x**, **P(x)** is false."

Example of Negating a Universal Quantifier

Consider the following statement −

$\forall x\ (x \geq 0)$

The negation of this statement would be −

$\neg \forall x\ (x \geq 0)$, which is equivalent to $\exists x\ (x < 0)$

This negation reads, "There exists an x such that $x < 0$" In the domain of real numbers, this is true, as there are negative numbers.

Example of Negating an Existential Quantifier

Consider the following statement −

$\exists x\ (x > 10)$

The negation of this statement would be −

$\neg \exists x\ (x > 10)$, which is equivalent to $\forall x\ (x \leq 10)$

This negation reads, "For all x, x is less than or equal to 10." If we are in the domain of natural numbers, this is false because numbers greater than 10 exist.

Implicit Quantifiers

In some cases, the quantifiers in a sentence are implied rather than explicitly stated. For example, when we say −

"If a shape is a square, then it is a rectangle"

Although this is not written with quantifiers, it is understood that we mean −

"For all shapes, if the shape is a square, then it is a rectangle"

Here, the universal quantifier "for all shapes" is **implicit**.

Similarly, when we use predicates, it is often assumed that the variables are universally quantified unless otherwise stated. For instance, when we define a predicate P(n) to mean "n is prime," we assume that n is universally quantified.

Using Predicates and Quantifiers in Proofs

Predicates and quantifiers are often used in formal proofs to express general statements about numbers, sets, or other mathematical objects. The combination of these tools allows for precision and clarity in mathematical reasoning.

Set - Definition
A set is an unordered collection of different elements. A set can be written explicitly by listing its elements using set bracket. If the order of the elements is changed or any element of a set is repeated, it does not make any changes in the set.

Some Example of Sets
- A set of all positive integers
- A set of all the planets in the solar system
- A set of all the states in India
- A set of all the lowercase letters of the alphabet

Representation of a Set
Sets can be represented in two ways −
- Roster or Tabular Form
- Set Builder Notation

Roster or Tabular Form
The set is represented by listing all the elements comprising it. The elements are enclosed within braces and separated by commas.
Example 1 − Set of vowels in English alphabet, $A=\{a,e,i,o,u\}$
Example 2 − Set of odd numbers less than 10, $B=\{1,3,5,7,9\}$

Set Builder Notation
The set is defined by specifying a property that elements of the set have in common. The set is described as $A=\{x:p(x)\}$

Example 1 − The set $\{a,e,i,o,u\}$
is written as −
$A=\{x$:x is a vowel in English alphabet$\}$

Example 2 − The set $\{1,3,5,7,9\}$
is written as −

$B=\{x:1\leq x<10$ *and* $(x\%2)\neq 0\}$
If an element x is a member of any set S, it is denoted by $x\in S$ and if an element y is not a member of set S, it is denoted by $y\notin S$
.
Example − If $S=\{1,1.2,1.7,2\}, 1\in S$ but $1.5\notin S$

Cardinality of a Set
Cardinality of a set S, denoted by $|S|$
, is the number of elements of the set. The number is also referred as the cardinal number. If a set has an infinite number of elements, its cardinality is ∞

.

Example − $|\{1,4,3,5\}|=4, |\{1,2,3,4,5,\ldots\}|=\infty$

If there are two sets X and Y,

- $|X|=|Y|$

denotes two sets X and Y having same cardinality. It occurs when the number of elements in X is exactly equal to the number of elements in Y. In this case, there exists a bijective function f from X to Y.

$|X|\leq|Y|$

denotes that set X_s cardinality is less than or equal to set Y_s cardinality. It occurs when number of elements in X is less than or equal to that of Y. Here, there exists an injective function f from X to Y.

$|X|<|Y|$

denotes that set X_s cardinality is less than set Y_s cardinality. It occurs when number of elements in X is less than that of Y. Here, the function f from X to Y is injective function but not bijective.

If $|X|\leq|Y|$ and $|X|\geq|Y|$ then $|X|=|Y|$

- . The sets X and Y are commonly referred as equivalent sets.

Types of Sets

Sets can be classified into many types. Some of which are finite, infinite, subset, universal, proper, singleton set, etc.

Finite Set

A set which contains a definite number of elements is called a finite set.

Example − $S=\{x|x\in N \text{ and } 70>x>50\}$

Infinite Set

A set which contains infinite number of elements is called an infinite set.

Example − $S=\{x|x\in N \text{ and } x>10\}$

Subset

A set X is a subset of set Y (Written as $X\subseteq Y$) if every element of X is an element of set Y.

Example 1 − Let, $X=\{1,2,3,4,5,6\}$ and $Y=\{1,2\}$. Here set Y is a subset of set X as all the elements of set Y is in set X. Hence, we can write $Y\subseteq X$

.

Example 2 − Let, $X=\{1,2,3\}$and $Y=\{1,2,3\}$. Here set Y is a subset (Not a proper subset) of set X as all the elements of set Y is in set X. Hence, we can write $Y\subseteq X$

.

Proper Subset

The term proper subset can be defined as subset of but not equal to. A Set X is a proper subset of set Y (Written as $X\subset Y$) if every element of X is an element of set Y and $|X|<|Y|$

.

Example − Let, $X=\{1,2,3,4,5,6\}$and $Y=\{1,2\}$. Here set $Y\subset X$ since all elements in Y are contained in X too and X has at least one element been more than set Y

.

Universal Set

It is a collection of all elements in a particular context or application. All the sets in that context or application are essentially subsets of this universal set. Universal sets are represented as U

.

Example − We may define U as the set of all animals on earth. In this case, set of all mammals is a subset of U, set of all fishes is a subset of U, set of all insects is a subset of U, and so on.

Empty Set or Null Set

An empty set contains no elements. It is denoted by ∅

. As the number of elements in an empty set is finite, empty set is a finite set. The cardinality of empty set or null set is zero.

Example − $S=\{x|x\in N \text{ and } 7<x<8\}=\emptyset$

Singleton Set or Unit Set

Singleton set or unit set contains only one element. A singleton set is denoted by $\{s\}$

.

Example − $S=\{x|x\in N, 7<x<9\}= \{8\}$

Equal Set

If two sets contain the same elements, they are said to be equal.

Example − If $A= \{1,2,6\}$and $B=\{6,1,2\}$

, they are equal as every element of set A is an element of set B and every element of set B is an element of set A.

Equivalent Set

If the cardinalities of two sets are same, they are called equivalent sets.

Example − If $A=\{1,2,6\}$ and $B=\{16,17,22\}$, they are equivalent as cardinality of A is equal to the cardinality of B. i.e. $|A|=|B|=3$

Overlapping Set

Two sets that have at least one common element are called overlapping sets. In case of overlapping sets −

- $n(A\cup B) =n(A)+n(B)-n(A\cap B)$
- $n(A\cup B) =n(A-B) +n(B-A) +n(A\cap B)$
- $n(A)=n(A-B) +n(A\cap B)$
- $n(B)=n(B-A) +n(A\cap B)$
-

Example − Let, $A= \{1,2,6\}$and $B=\{6,12,42\}$

. There is a common element 6, hence these sets are overlapping sets.

Disjoint Set

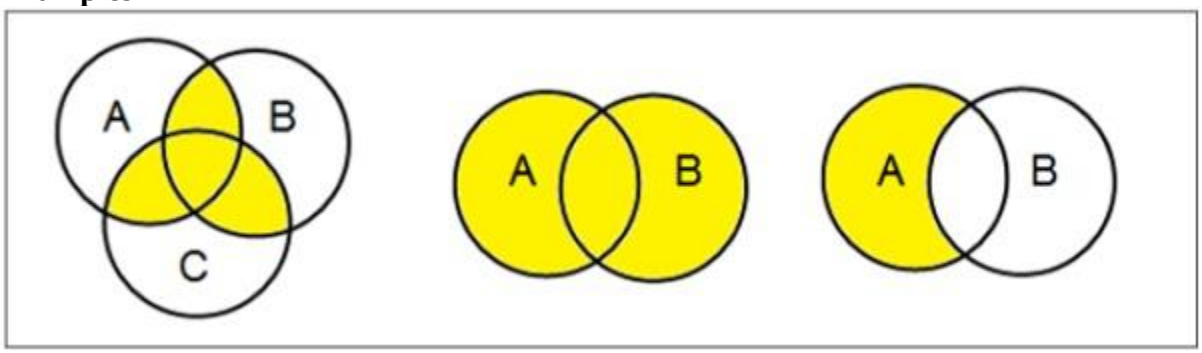
Two sets A and B are called disjoint sets if they do not have even one element in common. Therefore, disjoint sets have the following properties −

- $n(A \cap B) = \emptyset$
- $n(A \cup B) = n(A) + n(B)$

Example − Let, $A=\{1,2,6\}$ and $B=\{7,9,14\}$, there is not a single common element, hence these sets are overlapping sets.

Venn Diagrams
Venn diagram, invented in 1880 by John Venn, is a schematic diagram that shows all possible logical relations between different mathematical sets.

Examples



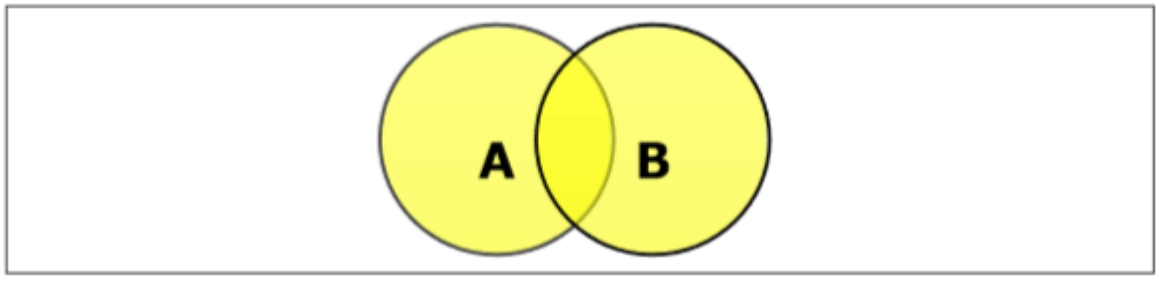
Set Operations
Set Operations include Set Union, Set Intersection, Set Difference, Complement of Set, and Cartesian Product.

Set Union
The union of sets A and B (denoted by $A \cup B$) is the set of elements which are in A, in B, or in both A and B. Hence, $A \cup B = \{x | x \in A \ OR \ x \in B\}$

.

Example − If $A = \{10,11,12,13\}$ and B = {13,14,15}, then $A \cup B = \{10,11,12,13,14,15\}$

. (The common element occurs only once)

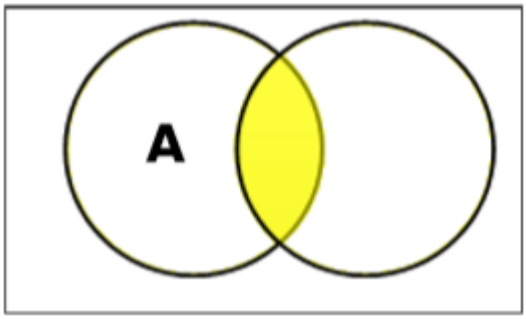


Set Intersection
The intersection of sets A and B (denoted by $A \cap B$) is the set of elements which are in both A and B. Hence, $A \cap B = \{x | x \in A \ AND \ x \in B\}$

.

Example − If $A = \{11,12,13\}$ and $B=\{13,14,15\}$, then $A \cap B = \{13\}$

.

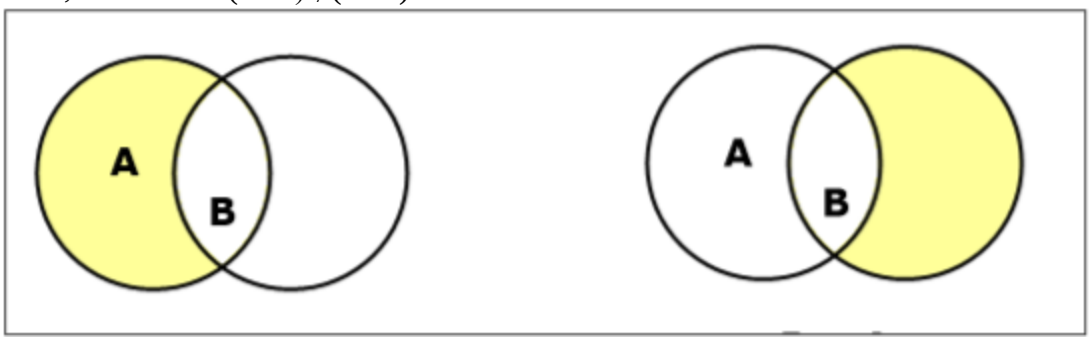


Set Difference/ Relative Complement

The set difference of sets A and B (denoted by AB) is the set of elements which are only in A but not in B. Hence, $A-B = \{x | x \in A \ AND \ x \notin B\}$

.

Example − If $A = \{10,11,12,13\}$and $B = \{13,14,15\}$, then $(A-B)=\{10,11,12\}$ and $(B-A)=\{14,15\}$. Here, we can see $(A-B) \neq (B-A)$



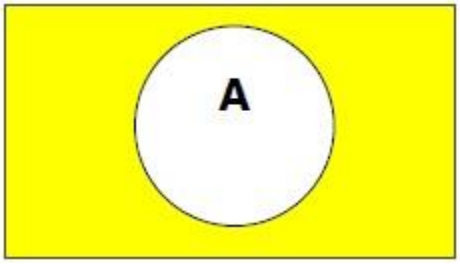
Complement of a Set

The complement of a set A (denoted by A) is the set of elements which are not in set A. Hence, $A' = \{x | x \notin A\}$

.

More specifically, $A' = (U-A)$ where U is a universal set which contains all objects.

Example − If $A = \{x | x \ belongstosetofoddintegers\}$then $A' = \{y | y \ doesnotbelongtosetofoddintegers\}$



Cartesian Product / Cross Product

The Cartesian product of n number of sets $A1, A2 \ldots An$

denoted as $A1 \times A2 \cdots \times An$ can be defined as all possible ordered pairs $(x1, x2 \ldots xn)$ where $x1 \in A1$, $x2 \in A2 \ldots xn \in An$

Example − If we take two sets $A=\{a,b\}$and $B=\{1,2\}$
,
The Cartesian product of A and B is written as − $A \times B=\{(a,1), (a,2), (b,1), (b,2)\}$
The Cartesian product of B and A is written as − $B \times A= \{(1, a), (1, b), (2, a), (2, b)\}$

Power Set
Power set of a set S is the set of all subsets of S including the empty set. The cardinality of a power set of a set S of cardinality n is 2^n . Power set is denoted as $P(S)$
.
Example −
For a set $S=\{a,b,c,d\}$
let us calculate the subsets −

- Subsets with 0 elements − $\{\emptyset\}$ (the empty set)
- Subsets with 1 element − $\{a\},\{b\},\{c\},\{d\}$
- Subsets with 2 elements − $\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}$
- Subsets with 3 elements − $\{a,b,c\},\{a,b,d\},\{a,c,d\},\{b,c,d\}$
- Subsets with 4 elements − $\{a,b,c,d\}$
-

Hence, $P(S)=$
$\{\{\emptyset\},\{a\},\{b\},\{c\},\{d\},\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\},\{a,b,c\},\{a,b,d\},\{a,c,d\},\{b,c,d\},\{a,b,c,d\}\}$
$|P(S)|=24=16$
Note − The power set of an empty set is also an empty set.
$|P(\{\emptyset\})|=20=1$

Sets and Notations in Discrete Mathematics
In Set Theory, we use elements inside sets as discrete elements. We can consider them as discrete elements in discrete mathematics. In several use-cases of discrete mathematics, we use sets. Sets are the foundational building blocks in discrete mathematics.

Definition of a Set
A **set** is an unordered collection of objects. These objects are called **elements** or **members** of the set. They can be anything: numbers, letters, other sets, or even abstract concepts.
In a set, **order does not matter** and **repetition is ignored** (an element is either in the set or not).

Following are some **examples of sets –**

- The set of all actors who have played The Doctor on Doctor Who
- The set of natural numbers between 1 and 10 inclusive

Set Notation
Here we will see the different types of set notations used in mathematics with their usecases and benefits.

Set Notation	Example and Descriptions
Element Notation	We use the symbol ∈ to denote that an object is an element of a set. **Example** − a ∈ {a, b, c} It is read as "a is an element of the set containing a, b, and c." Conversely, we use ∉ to denote that an object is not an element of a set. **Example** − d ∉ {a, b, c}
Set Builder Notation	Set builder notation is used to describe sets, especially when listing all elements is impractical. The general form is: {x : condition(s) that x must satisfy} **Example** − A = {x ∈ N : ∃n ∈ N(x = 2n)} This is read as "A is the set of all **x** in the natural numbers such that there exists some **n** in the natural numbers for which **x** is twice **n**." In simpler terms, A is the set of all even natural numbers. A simpler way to write this could be − **A = {x ∈ N : x is even}**

Special Sets

There are some special types of sets that play an important role in set theory

Special Sets	Descriptions
Empty Set	The empty set, denoted by ∅, is the set containing no elements.
Universal Set	The universal set, often denoted by U, is the set of all elements under consideration.
Common Number Sets	• N: The set of natural numbers (including 0) • Z: The set of integers • Q: The set of rational numbers • R: The set of real numbers Set Operations and Relations

Subsets

A subset can either be an ordinary subset or a proper subset.

Subset Types	Descriptions
Subset	A set A is a subset of set B (written A ⊆ B) if every element of A is also an element of B.
Proper Subset	A set A is a proper subset of set B (written A ⊂ B) if A ⊆ B and A ≠ B.

Set Operations

The operations on sets are given in the following table in short. We will see the operations with examples in a later chapter in detail.

Operations	Descriptions
Intersection	The intersection of sets A and B (written A ∩ B) is the set containing all elements that are in both A and B.
Union	The union of sets A and B (written A ∪ B) is the set containing all elements that are in A or B or both.
Cartesian Product	The Cartesian product of sets A and B (written A × B) is the set of all ordered pairs (a, b) where a ∈ A and b ∈ B.
Set Difference	The set difference of A and B (written A \ B) is the set containing all elements of A that are not in B.
Complement	The complement of set A (written A) is the set of all elements in the universal set that are not in A.

Some **other concepts** in sets are given below −

Operations	Descriptions
Power Set	The power set of A (written P(A)) is the set of all subsets of A.
Cardinality	The cardinality of a set A (written |A|) is the number of elements in A.

Example of Related to Sets

Let us take a look at some of the examples of these topics related to sets for a better understanding −

- **Describe the set {x: x + 3 ∈ N}** − This is the set of all numbers which, when 3 is added to them, result in a natural number. The set could be written as {-3, -2, -1, 0, 1, 2, ...}.
- **Describe the set {x ∈ N: x + 3 ∈ N}**: Solution − This is the set of all natural numbers which, when 3 is added to them, result in a natural number. So here we just have {0, 1, 2, 3 ...}.
- **Describe the set {x: x ∈ N ∨ -x ∈ N}** − This is the set of all integers (positive and negative whole numbers, written Z). In other words, {... , -2, -1, 0, 1, 2, ...}.
- **Describe the set {x: x ∈ N ∧ -x ∈ N}** − Here we want all numbers x such that x and -x are natural numbers. There is only one: 0. So we have the set {0}.
-

Set Theory Notation Table

All the symbols that are used in sets are given in the following table, for a quick reference –

Symbol	Meaning
{, }	Used to enclose elements of a set
:	Such that
∈	Is an element of
∉	Is not an element of
⊆	Is a subset of
⊂	Is a proper subset of
∩	Intersection
∪	Union
×	Cartesian product
\	Set difference
A	Complement of A
|A|	Cardinality of A
∅	Empty set
U	Universal set
N	Set of natural numbers
Z	Set of integers
Q	Set of rational numbers
R	Set of real numbers
P(A)	Power set of A

Relations

Relations may exist between objects of the same set or between objects of two or more sets.

Definition and Properties

A binary relation R from set x to y (written as xRy or $R(x,y)$) is a subset of the Cartesian product $x \times y$. If the ordered pair of G is reversed, the relation also changes.

Generally, an n-ary relation R between sets $A_1 \ldots$, *and* A_n is a subset of the n-ary product $A1 \times \cdots \times An$. The minimum cardinality of a relation R is Zero and maximum is $n2$ in this case. A binary relation R on a single set A is a subset of $A \times A$

.

For two distinct sets, A and B, having cardinalities m and n respectively, the maximum cardinality of a relation R from A to B is mn.

Domain and Range

If there are two sets A and B, and relation R have order pair (x, y), then −

- The **domain** of R, Dom(R), is the set $\{x|(x,y) \in R \text{ for some } y \text{ in } B\}$
- The **range** of R, Ran(R), is the set $\{y|(x,y) \in R \text{ for some } x \text{ in } A\}$

Examples

Let, $A = \{1,2,9\}$ and $B = \{1,3,7\}$

- Case 1 − If relation R is 'equal to' then $R = \{(1,1), (3,3)\}$

Dom(R) = {1,3}, $Ran(R) = \{1,3\}$

- Case 2 − If relation R is 'less than' then R= {(1,3), (1,7), (2,3), (2,7)}

Dom(R) = {1,2}, *Ran*(R)= {3,7}

- Case 3 − If relation R is 'greater than' then R= {(2,1), (9,1), (9,3), (9,7)}

Dom(R) = {2,9}, *Ran*(R)= {1,3,7}

Types of Relations

- The **Empty Relation** between sets X and Y, or on E, is the empty set ∅
- The **Full Relation** between sets X and Y is the set $X \times Y$
- The **Identity Relation** on set X is the set $\{(x,x)|x \in X\}$
- The Inverse Relation R' of a relation R is defined as − $R' = \{(b,a)|(a,b) \in R\}$

Example − If R= {(1,2), (2,3)} then R' will be {(2,1), (3,2)}

- A relation R on set A is called **Reflexive** if $\forall a \in A$ is related to a (aRa holds)

Example − The relation $R=\{(a,a),(b,b)\}$on set $X=\{a,b\}$

- is reflexive.
- A relation R on set A is called **Irreflexive** if no $a \in A$ is related to a (aRa does not hold).

Example − The relation $R=\{(a,b),(b,a)\}$on set $X=\{a,b\}$

- is irreflexive.
- A relation R on set A is called **Symmetric** if xRy mplies yRx, $\forall x \in A$ and $\forall y \in A$

.

Example − The relation R= {(1,2), (2,1), (3,2), (2,3)} on set A={1,2,3}

- is symmetric.
- A relation R on set A is called **Anti-Symmetric** if xRy and yRx implies $x=y \forall x \in A$ and $\forall y \in A$

.

Example − The relation $R=\{(x,y) \rightarrow N|x \leq y\}$is anti-symmetric since $x \leq y$ and $y \leq x$ implies $x=y$

- A relation R on set A is called **Transitive** if xRy and yRz implies $xRz, \forall x,y,z \in A$

.

Example − The relation R= {(1,2),(2,3),(1,3)}on set A={1,2,3}

- is transitive.
- A relation is an **Equivalence Relation** if it is reflexive, symmetric, and transitive.

Example − The relation R= {(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2), (1,3), (3,1)}
on set A= {1,2,3} is an equivalence relation since it is reflexive, symmetric, and transitive.

Set Operations

Union of Sets

The most basic operation on sets is **union**. The union of two sets combines all elements from both sets into one. We can say this mathematically as the union of sets A and B is written as (A

∪ B). This operation includes all elements that are in A, in B, or in both. It is important to note that duplicate elements are not counted twice.

Definition: The union of sets A and B is the set containing all elements that are either in A, in B, or in both.

Symbol − (A ∪ B)

Example − If A = {1, 2, 3} and B = {2, 3, 4}, then A ∪ B = {1, 2, 3, 4}

Intersection of Sets
The next widely used operation is the intersection of two sets. Intersection finds the common elements shared by both sets. It is denoted by (A ∩ B) and contains only the elements that exist in both A and B.

Definition: The intersection of sets A and B is the set containing all elements that are present in both A and B.

Symbol − (A ∩ B)

Example − If A = {1, 2, 3} and B = {2, 3, 4}, then A ∩ B = {2, 3}

Set Difference
The next operation is the difference of two sets. This is denoted as (A – B). Set difference gives the elements that are in A but not in B. It is also called the relative complement of B in A.

Definition: The difference of sets A and B is the set of all elements that belong to A but not to B.

Symbol − (A – B)

Example − If A = {1, 2, 3} and B = {2, 3, 4}, then A – B = {1}.

Complement of a Set
After union and intersection, another fundamental operation is complement. Remember the difference is a combination of these fundamental operations. The complement of a set **A**, denoted as $\mathbf{A^C}$ or **A′**. Complement contains all the elements not in **A**, assuming there is a universal set **U** that includes all possible elements under consideration.

Definition: The complement of set A consists of all elements in the universal set U that are not in A.

Symbol − A^C or A′
Example − If the universal set U = {1, 2, 3, 4, 5} and A = {1, 2, 3} then A^C = {4, 5}

Symmetric Difference

The symmetric difference between two sets A and B, which is denoted by (**A B**), is the set of elements that are in either of the sets but not in their intersection.
Definition: The symmetric difference is the set of elements that are in one of the sets but not in both.

Symbol − (A B)

Example − If A = {1, 2, 3} and B = {2, 3, 4} then A B = {1, 4}.

Cartesian Product
The Cartesian product of two sets A and B, denoted as (A × B). It is the set of all ordered pairs where the first element is from A and the second is from B.

Definition: The Cartesian product of sets A and B is the set of all ordered pairs (a, b) where a ∈ A and b ∈

Symbol − (A × B)

Example − If A = {1, 2} and B = {3, 4}, then A × B = {(1, 3), (1, 4), (2, 3), (2, 4)}

Examples of Set Operations
Here are a couple of examples to illustrate set operations in greater detail.

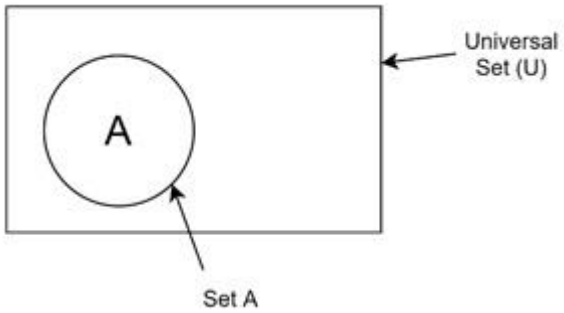
Example 1 − Let A = {1, 2, 3, 4, 5, 6}, B = {2, 4, 6}, C = {1, 2, 3}, and D = {7, 8, 9}

- A ∪ B = {1, 2, 3, 4, 5, 6} (since everything in B is already in A).
- A ∩ B = {2, 4, 6} (elements common to both sets).
- B ∩ C = {2} (the only element common to both B and C).
- A ∩ D = ∅ (no common elements between A and D).
- B ∪ C = {1, 2, 3, 4, 6} (union of B and C).
- A – B = {1, 3, 5} (elements in A but not in B).

Example 2 − Let A = {1, 2} and B = {3, 4, 5}

- A × B = {(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)}
- A × A = {(1, 1), (1, 2), (2, 1), (2, 2)}

Using Venn Diagrams
Sometimes we need to represent sets through diagrams. These are known as Venn diagrams. These diagrams are a visual tool for representing set operations.



In Venn diagrams, sets are represented by overlapping circles. The intersection, union, and other operations can be shown by shading the appropriate regions of the circles.

- **Union** − The union of two sets is shown by shading the entire area covered by both circles.
- **Intersection** − The intersection is represented by shading only the region where the circles overlap.
- **Difference** − The difference A – B is represented by shading the part of A that does not overlap with B.

Venn diagrams help visualize the relationships between sets and can simplify the process of understanding more complex operations.

Important Properties of Set Operations

Let us see some of the key properties of set operations −

Commutativity −

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

Associativity −

- $A \cup (B \cup C) = (A \cup B) \cup C$
- $A \cap (B \cap C) = (A \cap B) \cap C$

Distributivity −

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

De Morgan's Laws −

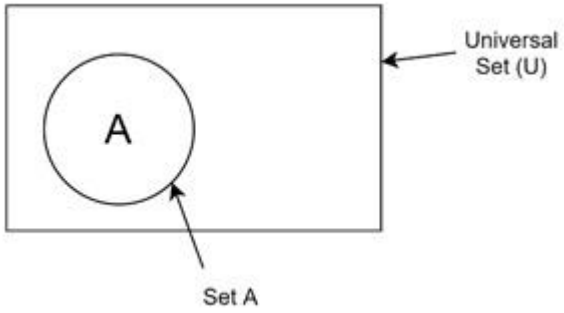
- $(A \cup B)^C = A^C \cap B^C$
- $(A \cap B)^C = A^C \cup B^C$

What are Venn Diagrams?

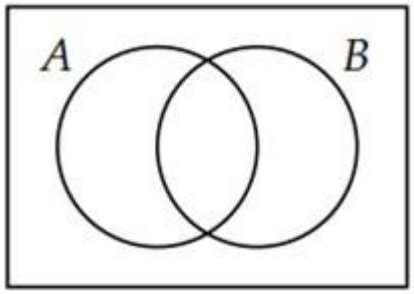
A Venn diagram is a tool to represent sets and their relationships with one another. Each set is typically represented as a circle, and the relationships between sets are shown by how these circles overlap.

- In a Venn diagram, each circle represents a set. When two or more sets overlap, the intersection (shared elements) is displayed in the overlapping region of the circles. Elements unique to each set are located in the non-overlapping parts.
- Often, the sets are contained within a rectangle that represents the **universal set**, denoted as **U**, which includes all the possible elements under consideration.

Take a look at the following Venn diagram. Here, we have the rectangle region as the Universe and there is a set **A** inside universe. All set are subset of the Universe.



Here, we get one example of overlapping sets. **A** and **B** are overlapping.

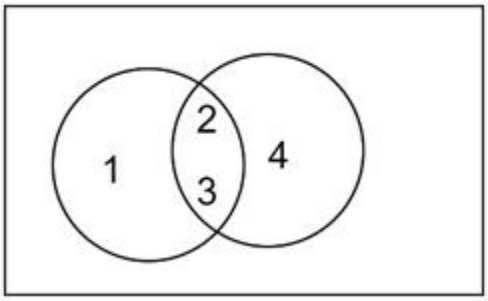


Example

Let us consider two sets −

- Set A = {1, 2, 3}
- Set B = {2, 3, 4}

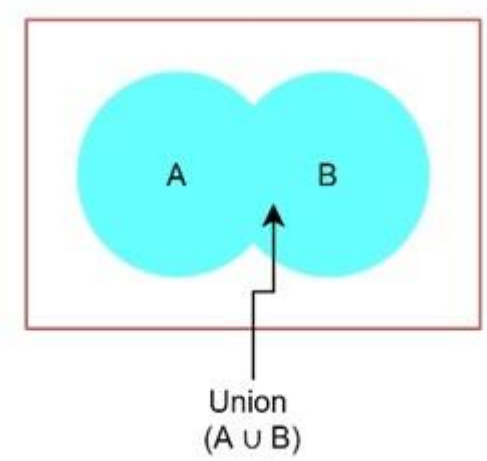
Using a Venn diagram, we draw two overlapping circles. The intersection (shared part) would be {2, 3} and the non-overlapping parts would be {1} for A and {4} for B.



Let us now check some **operations** that are usually done **on Venn Diagrams**.

Union of Sets

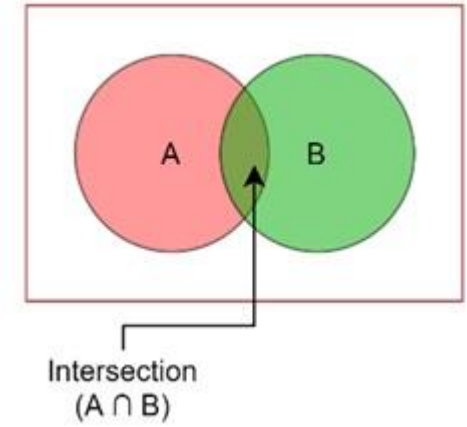
The **union** of two sets **A** and **B** denoted as (**A ∪ B**, is the set of elements that are in either **A**, **B**, or both. In a Venn diagram, the union is represented by shading the entire area covered by both circles.



In the Venn diagram, both circles would be completely shaded.

Intersection of Sets

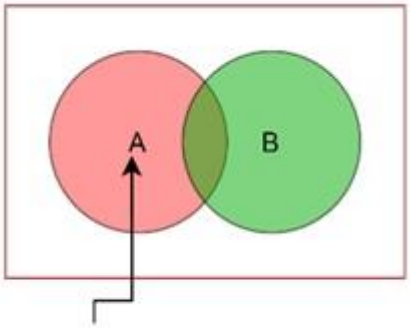
The intersection of two sets **A** and **B**, denoted by **A ∩ B**, is the set of elements common to both sets. In a Venn diagram, the intersection is shown by shading only the overlapping region of the circles.



In the Venn diagram, only the overlapping part of the two circles would be shaded.

Difference of Sets

The difference of two sets **A** and **B**, denoted by **A − B**, is the set of elements that are in **A** but not in B. In a Venn diagram, the difference is represented by shading only the part of **A** that does not overlap with **B**.



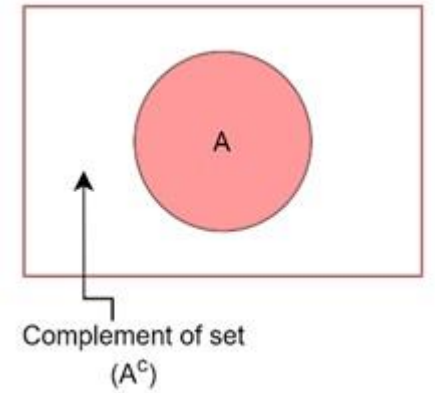
Set Difference (A-B)

In the Venn diagram, only the non-overlapping part would be shaded.

Complement of a Set

The complement of a set **A**, denoted by **A′** of $\mathbf{A^C}$, contains all the elements that are not in **A**. The complement depends on the universal set **U**, which includes all possible elements.

In a Venn diagram, the complement is represented by shading the area outside the circle of set **A**.

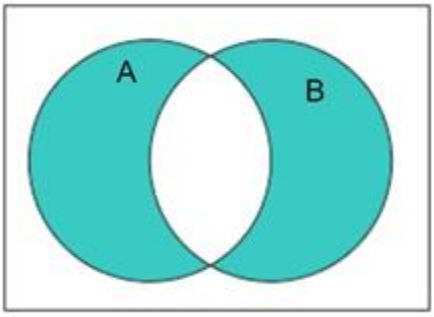


Complement of set (A^C)

In the Venn diagram, the area outside of the circle representing set **AAA** would be shaded.

Symmetric Difference

The **symmetric difference** between two sets **A** and **B**, denoted by **AB**, is the set of elements that are in either **A** or **B**, but not in both. In a Venn diagram, this is represented by shading the non-overlapping parts of both circles.



In the Venn diagram, the parts of **A** and **B** that do not overlap would be shaded.

Venn Diagrams with More Than Two Sets

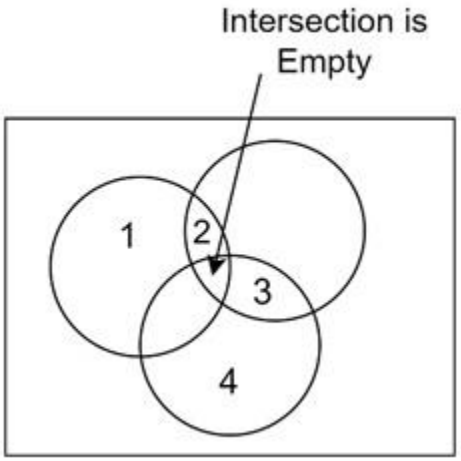
Venn diagrams are not limited to two sets. When we work with three sets or more, the diagrams become more complex, but the basic principles remain the same. We simply add more circles to represent additional sets, and their intersections and unions can be observed visually.

Example with Three Sets

Let us consider three sets −

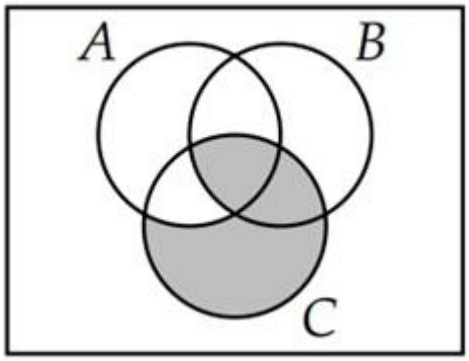
- A = {1, 2}
- B = {2, 3}
- C = {3, 4}

In its Venn diagram, the intersection of all three sets, **A ∩ B ∩ C**, would be empty (since no element is in all three sets).



The pairwise intersections, such as **A ∩ B = {2}** and **B ∩ C = {3}**, would be shown by overlapping regions of the circles.

Another complicated **example** could be: **(B ∩ C) U (C ∩ A^C)**, this could be represented as the following Venn diagram:



Functions

A **Function** assigns to each element of a set, exactly one element of a related set. Functions find their application in various fields like representation of the computational complexity of algorithms, counting objects, study of sequences and strings, to name a few.

Function - Definition

A function or mapping (Defined as $f: X \rightarrow Y$) is a relationship from elements of one set X to elements of another set Y (X and Y are non-empty sets). X is called Domain and Y is called Codomain of function f.

Function f is a relation on X and Y such that for each $x \in X$, there exists a unique $y \in Y$ such that $(x,y) \in R$. x is called pre-image and y is called image of function f.

A function can be one to one or many to one but not one to many.

Injective / One-to-one function

A function $f: A \rightarrow B$ is injective or one-to-one function if for every $b \in B$, there exists at most one $a \in A$ such that $f(s)=t$

.

This means a function **f** is injective if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$

.

Example

- $f: N \rightarrow N, f(x)=5x$ is injective.
- $f: N \rightarrow N, f(x)=x^2$ is injective.
- $f: R \rightarrow R, f(x)=x^2$ is not injective as $(-x)^2=x^2$

Surjective / Onto function

A function $f: A \rightarrow B$ is surjective (onto) if the image of f equals its range. Equivalently, for every $b \in B$, there exists some $a \in A$ such that $f(a)=b$. This means that for any y in B, there exists some x in A such that $y=f(x)$

.

Example

- $f: N \rightarrow N, f(x)=x+2$ is surjective.
- $f: R \rightarrow R, f(x)=x^2$ is not surjective since we cannot find a real number whose square is negative.

Bijective / One-to-one Correspondent

A function $f: A \rightarrow B$ is bijective or one-to-one correspondent if and only if **f** is both injective and surjective.

Problem

Prove that a function $f: R \rightarrow R$ defined by $f(x)=2x3$ is a bijective function.

Explanation − We have to prove this function is both injective and surjective.

If $f(x_1) = f(x_2)$, then $2x_13=2x_23$ and it implies that $x_1=x_2$

.
Hence, f is **injective**.
Here, $2x3=y$
So, $x=(y+5)/3$
which belongs to R and $f(x)=y$
.
Hence, f is **surjective**.
Since **f** is both **surjective** and **injective**, we can say **f** is **bijective**.

Inverse of a Function
The **inverse** of a one-to-one corresponding function f: $A\rightarrow B$, is the function g: $B\rightarrow A$, holding the following property −
$f(x)=y\Leftrightarrow g(y)=x$
The function f is called **invertible**, if its inverse function g exists.

Example
- A Function $f:Z\rightarrow Z, f(x)=x+5$, is invertible since it has the inverse function $g:Z\rightarrow Z, g(x)=x-5$
- A Function $f:Z\rightarrow Z, f(x)=x^2$ is not invertiable since this is not one-to-one as $(-x)^2=x^2$

Composition of Functions
Two functions f: $A\rightarrow B$ and g: $B\rightarrow C$ can be composed to give a composition gof. This is a function from A to C defined by $(gof)(x)=g(f(x))$

Example
Let $f(x)=x+2$ and $g(x)=2x+1$, find $(fog)(x)$ and $(gof)(x)$
.
Solution
$(fog)(x)=f(g(x)) =f(2x+1) =2x+1+2=2x+3$
$(gof)(x)=g(f(x)) =g(x+2) =2(x+2) +1=2x+5$

Hence, $(fog)(x)\neq(gof)(x)$

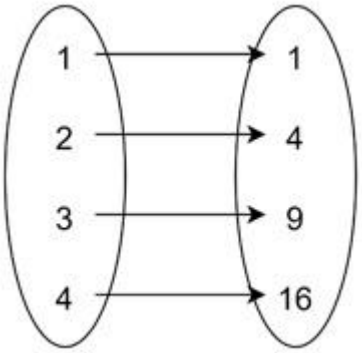
Some Facts about Composition
- If f and g are one-to-one then the function (gof) is also one-to-one.
- If f and g are onto then the function (gof) is also onto.
- Composition always holds associative property but does not hold commutative property.

Surjection, Injection and Bijection Function
Functions are nothing but mapping relationships between different sets. To fully understand functions, we need to understand three key concepts: surjection, injection, and bijection. In this chapter, we will see each of these function types with examples and see how they differ from one another.

Functions in Sets and Relations

A function is a rule that takes elements from one set (domain) and assigns them to elements in another set (codomain). Every element in the domain must map to exactly one element in the codomain. However, the reverse is not necessarily true. For example, imagine we assign every student in a class a locker. Each student gets exactly one locker, but some lockers may remain unused.

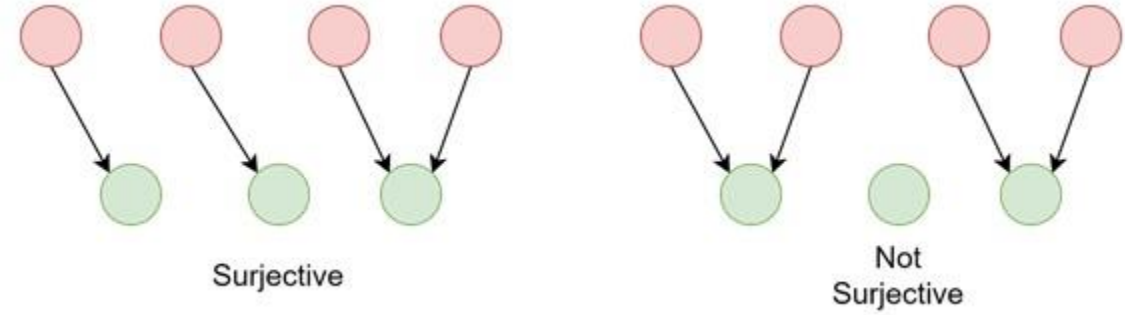
Here is another example. Take a look at the following mapping: Here, each number from left has a square on the right −



Surjection Functions That Are "Onto"

A surjective function, also known as an "onto" function. It is a function where every element in the codomain is the image of at least one element in the domain. In simpler words, there are no unused elements in the codomain.

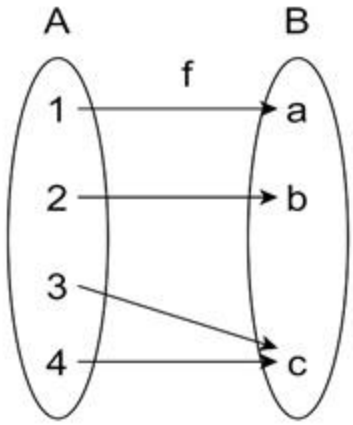
Formally, a function is surjective if every element of the codomain is the image of at least one element from the domain. Nothing in the codomain is left out.



Example of Surjection

Let us consider the function **f: {1, 2, 3} → {a, b, c}** defined by: **f (1) = a, f (2) = b, f(3) = c**

Here, every element in the codomain {a, b, c} has been "hit" by some element from the domain {1, 2, 3}. This means **f** is a surjective function because there are no elements left out in the codomain. The following mapping presents another **example** −



Example of Non-Surjective

Now, consider a function **g : {1, 2, 3} → {a, b, c}** where: **g(1) = a, g(2) = a, g(3) = c**.

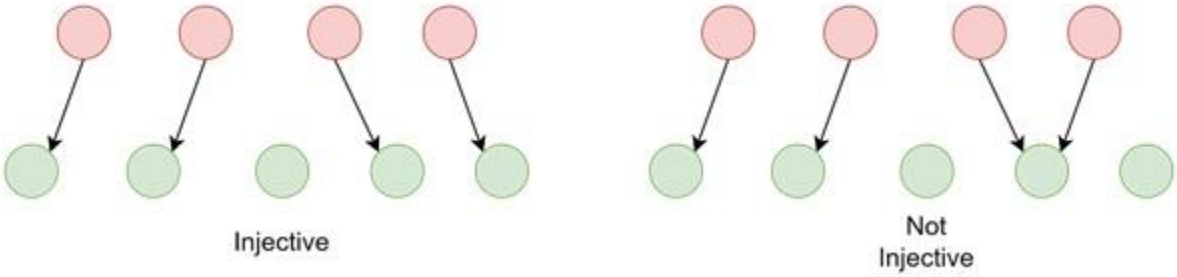
In this case, the element b from the codomain is never "hit" by any element from the domain. This means g is **not** surjective because something in the codomain is left out.

Surjections are important when we want to ensure that every possible output is covered. There is no leaving "gaps."

Injection Functions That Are "One-to-One"

On the other hand, the injective function, or "one-to-one" function, shows that no two different elements in the domain map to the same element in the codomain. This means every element in the codomain is the image of at most *one* element in the domain.

Formally, a function is injective if every element of the codomain is the image of at most one element from the domain. There are no "repeated" images in the codomain.



Example of Injection

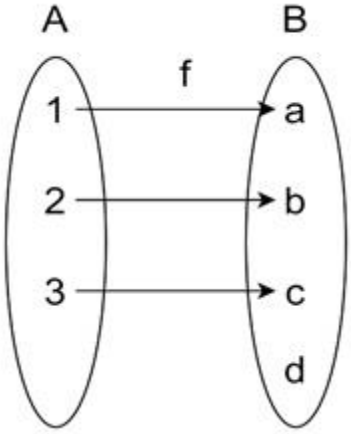
Let us consider **f : Z → Z** (where **Z** represents all integers) is defined by: **f(n) = 3n**

For every integer **n**, **f(n)** produces a unique result. No two different integers will give the same value after being multiplied by 3. Therefore, this function is injective. It does not "reuse" codomain elements for different domain inputs.

Example of Non-Injective

Consider a function **h : {1, 2, 3} → {a, b}** defined by: **h(1) = a, h(2) = a, h(3) = b**

In this case, both h(1) and h(2) are mapped to a. Since two different domain elements are assigned to the same codomain element, h is **not** injective. Injection fails when there is "overlap" in the codomain.

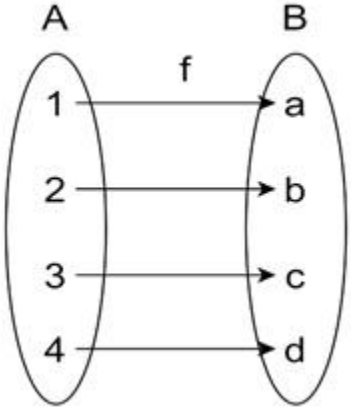


Bijection The Perfect Function

Next comes the bijection functions. It is both injective and surjective. In other words, every element in the domain maps to a unique element in the codomain, and every element in the codomain is covered. This makes bijections "perfect" functions because they pair every element in the domain with one (and only one) element in the codomain, with no repeats and no leftovers. Formally, a function is bijective if it is both injective and surjective, meaning it's a one-to-one correspondence between the domain and codomain.

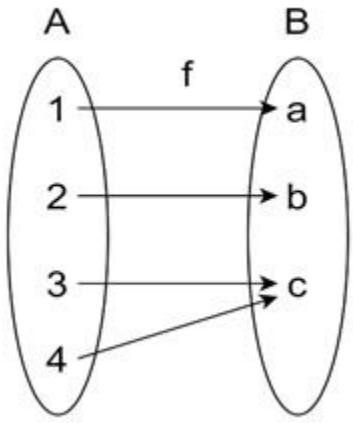
Example of Bijection

Consider the function **f : {1, 2, 3} → {a, b, c}**, defined as: **f(1) = a, f(2) = b, f(3) = c**.
This is a bijection because every element in the domain has a unique match in the codomain, and every codomain element is used. There are no repeated values and no leftover elements.



Example of Non-Bijective Function

If we modify **f** such that **f(2) = a**, then the function is no longer bijective. It might still be surjective (depending on the other values), but it is no longer injective because two elements in the domain share the same codomain value.



Combining Concepts: Injective vs Surjective

We understood the idea of injective and surjective. But these are essential to understand that injective and surjective functions are not opposites.

A function can be injective but not surjective; surjective but not injective; both; or neither.

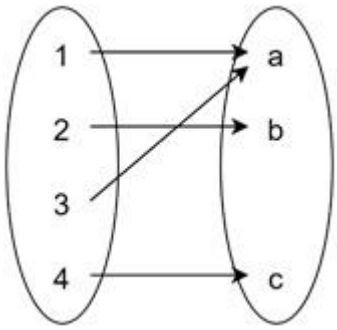
- A function can be injective but miss out some elements in the codomain, making it not surjective.
- A function can be surjective but reuse elements of the codomain, making it not injective.
- A bijection is both injective and surjective, satisfying both properties.

What is the Image of a Function?

The image of a function is the set of all outputs that the function produces when applied to elements from the domain. More specifically, it is the set of all elements in the codomain that are "hit" or "mapped to" by some element in the domain.

Example of Image

Consider a function **f: {1, 2, 3, 4} → {a, b, c}** defined as: **f(1) = a**, **f(2) = b**, **f(3) = a**, **f(4) = c**

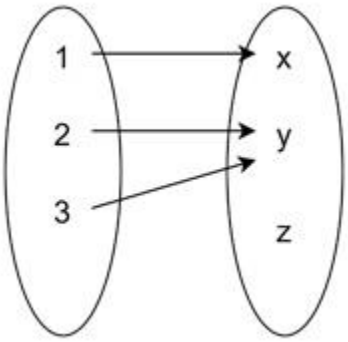


Here, the image of f is the set of all elements in the codomain that are mapped by the function. So, in this case, the image is: **Image (f) = {a, b, c}**

The image includes every codomain element that is assigned to at least one domain element. In our example, every element in the codomain is part of the image because every element of {a, b, c} has at least one input from the domain.

Example of Partial Image

Now, imagine a different function **g: {1, 2, 3} → {x, y, z}** defined as: **g (1) = x**, **g(2) = y**, **g(3) = y**



Here, the image is: **Image (g) = {x, y}**

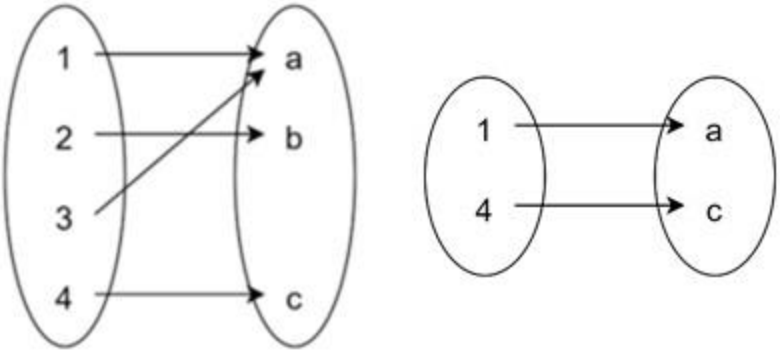
Notice that **z** is not part of the image because no element from the domain maps to **z**. This happens when some elements of the codomain are left "unused."

Subset Image

Another concept of image is the subset image. Concept of image can also apply to a subset of the domain. If we want to know the image of only part of the domain, we can restrict the function to that subset.

Example of Subset Image

Consider the same function **f: {1, 2, 3, 4} → {a, b, c}**, defined as before. Now, we want to find the image of the subset {1, 4}: **Image of {1, 4} = {f (1), f (4)} = {a, c}**



In this case, we are only concerned with the elements in the codomain that correspond to inputs from the subset {1, 4}. The image is simply {a, c}.

Inverse-Image of a Function

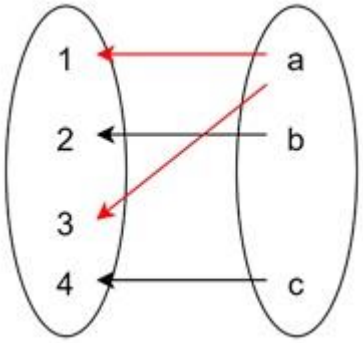
Let us now understand the inverse-image (also called the **preimage**). Given an element in the codomain, which elements in the domain map to it?

Instead of starting with elements from the domain and mapping to the codomain (as with the image), the inverse-image starts with elements in the codomain and traces back to their corresponding elements in the domain.

Example of Inverse-Image

Let us consider the same function **f: {1, 2, 3, 4} → {a, b, c}** defined as: **f (1) = a, f(2) = b, f(3) = a, f(4) = c**

Now, suppose we want to find the inverse-image of the element a in the codomain. The inverse-image of a is the set of all elements in the domain that map to a. From the definition of the function, we see that **f (1) = a** and **f(3) = a**, so: $\mathbf{f^{-1}(a) = \{1, 3\}}$

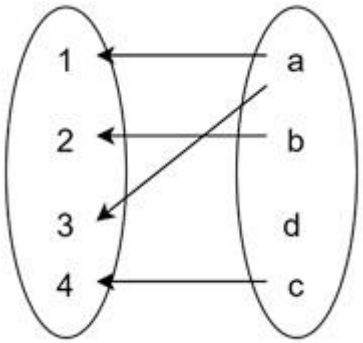


The inverse-image of **a** includes both 1 and 3, because both of these domain elements map to **a**.

Empty Inverse-Image

Sometimes, there may be codomain elements that no domain elements map to. In this case, the inverse-image of those elements is the empty set.

For example, if we wanted to find $\mathbf{f^{-1}(d)}$ for our function f, we would notice that no domain element maps to **d**. Therefore, the inverse-image of **d** is: $\mathbf{f^{-1}(d) = \emptyset}$



This simply means that **d** is not mapped by any element from the domain.

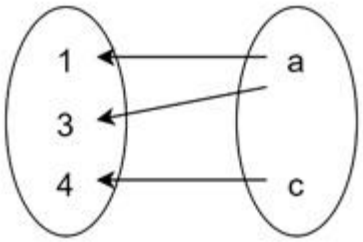
Inverse-Image of a Subset

Just like with the image of a subset of the domain, we can also find the inverse-image of a subset of the codomain. In this case, we look for all domain elements that map to *any* element in the codomain subset.

Example of Inverse-Image of a Subset

Again, using our function **f : {1, 2, 3, 4} → {a, b, c}**, let us find the inverse-image of the subset {a, c} in the codomain: $\mathbf{f^{-1}(\{a, c\}) = \{1, 3, 4\}}$

This is because: **f(1) = a**, **f(3) = a**, **f(4) = c**



Therefore, the inverse-image of {a, c} includes all domain elements that map to either **a** or **c**.

Inverse-Images: Points to Note

The inverse-image of an element in the codomain is always a set, not a single element. This is because multiple domain elements can map to the same codomain element, so the inverse-image might contain more than one element.

The inverse-image only refers to which domain elements map to a given codomain element or set of elements. It is not the same as an inverse function. Inverse functions only exist when the function is a bijection (one-to-one and onto), but inverse-images exist for all types of functions.

Combining Image and Inverse-Image

The concepts of image and inverse-image are closely related, and they are used to understand how a function behaves. The image gives us insight into which codomain elements are "hit" by the function, while the inverse-image tells us where those hits came from in the domain.

In general, the image helps us see the "forward" movement of a function, from the domain to the codomain. The inverse-image allows us to trace backwards, from the codomain to the domain, and identify which domain elements correspond to certain codomain elements.

Proof by Contrapositive in Discrete Mathematics

In Discrete mathematics, we use logics to deduct and prove in several ways. Another way of proving logics is using the techniques with the **proof by contrapositive**. This method demonstrates the truth of an implication by proving the equivalent contrapositive statement instead. It is a useful and often simpler way to verify a claim when direct proof methods might be cumbersome or unclear.

What is Proof by Contrapositive?

To understand the concept of proof by contrapositive, we must cover the basics of **implications**. Let us first recall what an implication is. In logical terms, an implication is a statement of the form "**if P, then Q**", which we write as −

$P \rightarrow Q$

Here, P is called the hypothesis (or premise), and Q is the conclusion.

The contrapositive of an implication reverses and negates both the hypothesis and the conclusion. So, the **contrapositive** of the statement $\mathbf{P \rightarrow Q}$ is −

$\neg Q \rightarrow \neg P$

This means that instead of proving "if P, then Q", we prove "if not Q, then not P". An important fact is that the contrapositive is logically equivalent to the original statement. In other words, if the contrapositive is true, the original implication must also be true.

Why Use Contrapositive Proofs?

Sometimes, it is difficult to prove a statement directly. The relationship between P and Q might be too complex or unclear when handling this. However, the contrapositive may offer a simpler or more intuitive way out. Giving the contrapositive can sometimes break down the problem into smaller, more manageable parts.

To get the idea clear on how a proof by contrapositive works, let us look at an example from the idea of even and odd numbers.

Example 1: Proving the Statement with Contrapositive

Consider the statement: **"For all integers n, if n^2 is even, then n is even"**.

We want to prove this implication. Proving it directly would mean starting with the assumption that n^2 is even and trying to show that n itself is even. While possible, this approach might not feel straightforward. Instead, let us prove its contrapositive:

The contrapositive of the given statement is −

"If n is odd, then n^2 is odd"

Proving this contrapositive will be enough to confirm the truth of the original implication. So, we start by assuming that **n** is odd and show that $\mathbf{n^2}$ must also be odd.

Step 1: Assume the negation of the conclusion

We start with assuming that **n** is odd. If **n** is odd, we can write **n** in the form −

$$n=2k+1$$

where **k** is an integer. This is the standard way to represent odd numbers because they differ from the nearest even number by 1.

Step 2: Square the assumption

Next, we need to find $\mathbf{n^2}$, the square of an odd number. Using the expression for **n**, we calculate −

$$n^2=(2k+1)^2$$

Expanding the square −

$$n^2=4k^2+4k+1$$

Step 3: Analyze the result

We can factor this expression −

$$n^2=2(2k^2+2k)+1$$

Notice that the expression $\mathbf{2(2k^2 + 2k)}$ is clearly an even number since it is divisible by 2. Therefore, $\mathbf{n^2}$ is of the form "even number + 1", which means that $\mathbf{n^2}$ is odd.

Step 4: Conclusion

Thus, we have shown that if **n** is odd, then $\mathbf{n^2}$ must also be odd. Since we have successfully proven the contrapositive, we can now conclude that the original statement is true −

"For all integers n, if n^2, then n is even."

Let us now see another effective example of Proof by Contrapositive.

Example 2: Divisibility and Contrapositive

Let us prove the following statement −

"For all integers a and b, if a × b is odd, then both a and b are odd."

Again, a direct proof could be tricky, but the contrapositive offers a simpler approach. The contrapositive of this statement is −

"If either a or b is even, then a × b is even."

This is much easier to prove. We assume that one of the numbers, say **a**, is even. If **a** is even, then we can write **a = 2k** for some integer **k**. Now, **a × b = (2k) × b = 2(k × b)**, which is clearly an even number because it is divisible by 2. Hence, we can conclude this is contrapositive, and by extension, the original statement is True.

Proof by Contradiction in Discrete Mathematics

Contradiction means negating a statement or when something false we care about. Proof by Contradiction is one of the most powerful methods used in discrete mathematics, especially when we are working on statements that are difficult to prove directly. The idea of this method lies in its simplicity; it assumes the opposite of what we are trying to prove, and shows how it leads to a contradiction.

In this chapter, we will explain Proof by Contraction in detail, discussing its basic framework and providing practical examples for a better understanding of this concept.

What is Proof by Contradiction?

Proof by Contradiction operates on the following logical principle: to prove a statement **P**, assume that **P** is false (i.e., assume **¬P**, the negation of **P** is true), and then show that this assumption leads to a contradiction. The contradiction arises when two mutually exclusive facts or statements are derived. Since the assumption **¬P** leads to a contradiction, the original statement **P** must be true.

The basic structure of a proof by contradiction follows these steps −

- **Assume the negation** of the statement we are trying to prove.
- **Develop the argument**, using logical steps and known facts.
- **Arrive at a contradiction**, a situation where a logical inconsistency arises.
- **Conclude** that the original statement must be true because the assumption that it was false leads to an impossible scenario.

Proof by Contradiction is useful when we are working with universal statements, the theorems involving inequalities, or proving the existence of something that cannot be easily demonstrated by construction.

Let us now understand the concept with the help of a set of examples.

Example 1: Proving √2 is Irrational

One of the most famous examples of proof by contradiction is the proof that √2 is irrational. The statement to be proved is −

"The square root of 2 is irrational."

Step 1: Assume the negation
Assume the opposite, i.e., that √2 is rational. We know by definition of rationality, this means that −
2–√=ab
where **a** and **b** are integers, and the fraction is in its simplest form (i.e., **a** and **b** have no common divisors other than 1).

Step 2: Develop the argument
By squaring both sides, we get −
$2=a^2b^2$ which implies that $-a^2=2b^2$
This equation tells us that $\mathbf{a^2}$ is an even number (because it is equal to $\mathbf{2b^2}$). From this, it follows that A itself must also be even (because the square of an odd number is odd, and the square of an even number is even). So, we can write −
a=2k
for some integer **k**. Substituting this into the equation $\mathbf{a^2 = 2b^2}$ we get −
$(2k)^2=2b^2$
$\Rightarrow 4k^2=2b^2$
$\Rightarrow b^2=2k^2$
Thus, $\mathbf{b^2}$ is also even, meaning that b is even.

Step 3: Arrive at the contradiction
At this point, we have an idea that that both **a** and **b** are even. But this contradicts our original assumption that ab
is in its simplest form (because if both **A** and **B** are even, then they share at least 2 as a common divisor). Therefore, our assumption that √2 is rational must be False.

Step 4: Conclude
Since the assumption led to a contradiction, we conclude that √2 is irrational.

Example 2: No Integer Solutions for $\mathbf{x^2 = 4y + 2}$
Let us consider another example of a quadratic equation,
"Prove that there are no integers x and y such that $x^2 = 4y + 2$"
Step 1: Assume the negation
Assume that there are integers x and y such that −
$x^2=4y+2$
Step 2: Develop the argument
We can rewrite this equation as −
$x^2=2(2y+1)$
This shows that $\mathbf{x^2}$ is an even number. If $\mathbf{x^2}$ is even, then **x** must also be even. This is because the square of an odd number is odd).
So, let **x = 2k** for some integer **k**. Substituting this into the equation, we get −
(2k)2=2(2y+1)
⇒4k2=2(2y+1)
⇒2k2=2y+1

Step 3: Arrive at the contradiction
The equation $2k^2 = 2y + 1$ implies that the left-hand side is even (since $2k^2$ is divisible by 2), but the right-hand side is odd (since $2y + 1$ is an odd number). This is a contradiction because an even number cannot equal an odd number.
Step 4: Conclude
Thus, the assumption that there exist integers **x** and **y** satisfying $x^2 = 4y + 2$ leads to a contradiction. So, no such integers exist.

Example 3: The Pigeonhole Principle
Let us see our third example, which is one of the most famous problem in discrete mathematics. The Problem of Pigeonhole Principle. It states that if more than n objects are placed into n containers, at least one container must hold more than one object. Here, how we can prove it using contradiction:
"If more than **n** pigeons fly into **n** pigeonholes, at least one pigeonhole must contain more than one pigeon."

Step 1: Assume the negation
Assume the opposite, that no pigeonhole contains more than one pigeon. In other words, each pigeonhole contains at most one pigeon.

Step 2: Develop the argument
If each pigeonhole contains at most one pigeon, then the total number of pigeons that can fit into the n pigeonholes is at most n (Each pigeonhole holds no more than one pigeon).
Step 3: Arrive at the contradiction
We have assumed that there are more than **n** pigeons. This gives a contradiction, because if there are more pigeons than pigeonholes, at least one pigeonhole must contain more than one pigeon.

Step 4: Conclude
Thus, the assumption that no pigeonhole contains more than one pigeon leads to a contradiction. Therefore, at least one pigeonhole must contain more than one pigeon. There are some other examples where we use this method of contradiction to prove something. In automata theory, a special branch of discrete math and computation, we use the concept called **pumping lemma** for **regular languages** and **contextfree languages**. For them as well, we use the method of contradiction to prove some expression is not falling under regular language or context free language.

Proof by Cases in Discrete Mathematics
What is Proof by Cases?
Proof by cases is simpler than other methods. The goal is to prove a statement P, but instead of proving **P** directly, we divide the problem into several cases, Q_1, Q_2, ..., Q_n. For each case, we proved that the statement **P** holds. Since one of the cases must always be true. So proofing the statement for all cases guarantees that **P** is universally true.
The structure of a Proof by Cases is as follows −

- **Identify all the cases** − These are the different conditions or scenarios under which the problem can occur.

- **Prove each case** − For each individual case, show that the statement holds.
- **Conclude** − Since one of the cases must be true, the overall statement is true.

This method is interesting and quite useful when the problem involves logical "or" conditions, or when a value can take different forms, such as being even or odd, positive or negative.

Why Use Proof by Cases?

Proof by Cases is helpful when the problem is complex and cannot be easily addressed with a single argument. Sometimes, different conditions lead to different approaches. For example, in number theory, a number can be even or odd, and proving the same statement for both conditions may require separate reasoning.

After breaking down the problem, Proof by Cases provides clarity and ensures that every possible situation is covered. This method is exhaustive, and it ensures there are no gaps in the proof.

Example 1: Proving $n^3 - n$ is Even for Any Integer n

Let us consider an example, for any integer **n**, the expression $\mathbf{n^3 - n}$ is always even.

Step 1: Identify the cases

The key observation here is that every integer n is either even or odd. So that, we can divide the problem into two cases −

- **Case 1** − **n** is even
- **Case 2** − **n** is odd

Step 2: Prove Each Case

Case 1: n is even − If **n** is even, we can write it like **n = 2k** for some integer **k**. Then substituting **n = 2k** into the expression $\mathbf{n^3 - n}$, we get −

$n^3 - n = (2k)^3 - 2k = 8k^3 - 2k = 2(4k^3 - k)$

Since $\mathbf{4k^3 - k}$ is an integer, the expression is divisible by 2, which means that $\mathbf{n^3 - n}$ is even when **n** is even.

Case 2: n is odd − If **n** is odd, we can write **n = 2k + 1** for some integer **k**. Substituting **n = 2k + 1** into $\mathbf{n^3 - n}$, we get:

$n^3 - n = (2k+1)^3 - (2k+1)$

Expanding the cubic term, we get −

$(2k+1)^3 = 8k^3 + 12k^2 + 6k + 1$

Thus,

$n^3 - n = (8k^3 + 12k^2 + 6k + 1) - (2k+1) = 8k^3 + 12k^2 + 4k$

Factoring out 2,

$n^3 - n = 2(4k^3 + 6k^2 + 2k)$

Again, the expression is divisible by 2, which means that $\mathbf{n^3 - n}$ is even when **n** is odd.

Step 3: Conclude

Since $\mathbf{n^3 - n}$ is even in both cases (here **n** is even or odd), we conclude that for any integer **n**, the expression $\mathbf{n^3 - n}$ is always even.

Example 2: Sum of Odd Numbers
Let us see another example. This one is another interesting problem. Prove the statement −
"If a + b is odd, then a is odd or b is odd."

Step 1: Identify the cases
We can handle this by considering the two possible scenarios for A and B:

- **Case 1** − Both **a** and **b** are even.
- **Case 2** − Both **a** and **b** are odd.

Step 2: Prove Each Case
Case 1: Both a and b are even − If both **a** and **b** are even, then we can write **a = 2m** and **b = 2n** for some integers **m** and **n**. Adding them together −
$a+b=2m+2n=2(m+n)$
Since **a + b** is divisible by 2, the sum is even, contradicting the assumption that **a + b** is odd. Therefore, this case does not hold.

Case 2: Both a and b are odd – If both a and b are odd, we can write **a = 2m + 1** and **b = 2n + 1** for some integers **m** and **n**. Adding them together −
$a+b=(2m+1)+(2n+1)=2(m+n+1)$
This sum is again even, which contradicts the assumption that **a + b** is odd. Therefore, this case also does not hold.

Step 3: Conclude
We have shown that if **a + b** is odd, then at least one of **a** or **b** must be odd, as the statement requires.

Example 3: Proof by Cases in Geometry
Let us see the third example in geometry. Consider the following statement −
"For any triangle, the sum of the interior angles is 180 degrees."

Step 1: Identify the cases
We can consider different types of triangles:

- **Case 1** − The triangle is acute (all angles are less than 90 degrees).
- **Case 2** − The triangle is right (one angle is 90 degrees).
- **Case 3** − The triangle is obtuse (one angle is greater than 90 degrees).

Step 2: Prove each case

Case 1: Acute Triangle − In an acute triangle, all angles are less than 90 degree. Regardless of the specific angle measures, the geometric properties of triangles ensure that the sum of the angles always equal to 180 degrees.

Case 2: Right Triangle − In a right triangle, one angle is exactly 90 degrees. The other two angles must sum to 90 degrees, ensuring that the total sum of the interior angles is 180 degrees.

Case 3: Obtuse Triangle − In an obtuse triangle, one angle is greater than 90 degrees, and the other two angles must be the sum to less than 90 degrees. Again, the total sum of the angles in the triangle will always be 180 degrees.

Step 3: Conclude
Since the interior angles of a triangle sum to 180 degrees in all possible cases, the statement is universally true for all triangles.

What is Sequence?

It is a set of numbers in a definite order according to some definite rule (or rules). Each number of the set is called a term of the sequence and its length is the number of terms in it. We can write the sequence as: $\{a_n\}_{n=1}^{\infty}$ *or* a_n. A finite sequence is generally described by a_1, a_2, a_3.... a_n, and an infinite sequence is described by a_1, a_2, a_3.... to infinity. A sequence $\{a_n\}$ has the limit L and we write:

$$\lim_{n\to\infty} a_n = L \text{ or } a_{n0}^{\infty} \text{ as } n \to \infty$$

For example:

- 2, 4, 6, 8 ...., 20 is a finite sequence obtained by adding 2 to the previous number.
- 10, 6, 2, -2, ... is an infinite sequence obtained by subtracting 4 from the previous number.

If the terms of a sequence can be described by a formula, then the sequence is called a **progression**.
1, 1, 2, 3, 5, 8, 13, ..., is a progression called the Fibonacci sequence in which each term is the sum of the previous two numbers.

Types of Sequences

1. **Arithmetic Sequence (AP)**
    - Each term differs from the previous one by a fixed number (common difference d).
    - Formula:

$a_n = a_1 + (n-1)\, d$

Example: 2,5,8, 11… (d=3)

Geometric Sequence (GP)

- Each term is obtained by multiplying the previous one by a fixed ratio r.
- Formula:

$a_n = a_1 . r^{n-1}$

Example: 3,6,12, 24…(r=2)

Fibonacci Sequence

- Defined recursively

  $F_1 = 1$, $F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$

- Sequence: 1,1,2,3,5,8, 13….,

Summation (Σ Notation)

Definition

The summation symbol ∑ represents the sum of terms of a sequence.

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \ldots a_n$$

- **Lower limit (m):** starting index.
- **Upper limit (n):** ending index.
- **General term:** a_k.

Examples

1. $\sum_{k=1}^{4} k = 1 + 2 + 3 + 4 = 10.$

2. $\sum_{k=1}^{5} (2k) = 2 + 4 + 6 + 8 + 10 = 30.$

Important Summation Formulas

1. *Sum of first n natural numbers* :

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

2. *Sum of Squares*:

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

3. *Sum of Cubes*:

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Series (Special Summations)

Arithmetic Series

Sum of first n terms of an AP:

$$S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}[2a_1 + (n-1)d]$$

Example

Find the sum of the first 20 terms of 5, 8, 11, …

$a_1 = 5$, d = 3, n =20

$$S_{20} = \frac{20}{2}[2(5) + (20-1)(3)] = 10[10 + 57] = 670$$

Geometric Series

Sum of first n terms of a GP:

$$S_n = a_1 \frac{1-r^2}{1-r}, (r \neq 1)$$

Example:

Find the sum of first 6 terms of 3, 6, 12, 24, …

a1 = 3, r= 2, n = 6

$$S_6 = 3.\frac{1-2^6}{1-2} = 3.\frac{1-64}{-1} = 3.63 = 189$$

Infinite Geometric Series

If |r| < 1, the infinite GP converges:

$$S_\infty = \frac{a_1}{1-r}$$

Example:

3 + 1.5 + 0.75 +…

Here $a_1 = 3, r = \frac{1}{2}$

$S\infty = \frac{3}{1-\frac{1}{2}} = 6$

Property of Summations

- **Linearity:**
  $\sum_{k=1}^{n}(a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$

- **Constant factor:**
  $$\sum_{k=1}^{n} c.a_k = c\sum_{k=1}^{n} a_k$$

Principle of Mathematical Induction

Mathematical induction is a concept in mathematics that is used to prove various mathematical statements and theorems. The principle of mathematical induction is sometimes referred to as PMI. It is a technique used to prove basic theorems in mathematics that involve solving up to n finite natural terms.

Suppose P(n) is a statement for n natural numbers, then it can be proved using the Principle of Mathematical Induction.

- We will prove for P (1), then let P(k) be true then prove for P(k+1).
- If P(k+1) holds, then we say that P(n) is true by the principle of mathematical induction.

Principle of Mathematical Induction Statement

Any statement P(n), which is for "n" a natural number, can be proved using the Principle of Mathematical Induction by following the steps below,

Step 1: Verify if the statement is true for trivial cases (**n = 1)** i.e. check if P(1) is true.
Step 2: Assume that the statement is true for n = k for some $k \geq 1$ i.e. P(k) is true.
Step 3: If the truth of P(k) implies the truth of P(k + 1), then the statement P(n) is true for all **$n \geq 1$**.

The first statement is a fact, and if it is not possible for all P(n) to hold at n = 1, then these statements are true for some other values of n, say n = 2, n = 3, and others.

If the statement is true for P(k), then if P(k+1) is proven to be true, then we say that P(n) is true for all n belonging to Natural Numbers (N)

Mathematical Induction Solution and Proof

Consider a statement P(n), where n is a natural number. Then, to determine the validity of P(n) for every n, use the following principle:

- **Base Step:** Prove P(k) is true for k =1

- **Assumption Step:** Assume that the given statement P (n) is also true for n = k, whether k is any positive integer.
- **Induction Step:** Prove P(k+1) is true using basic mathematical properties.

If the above three steps are proved, then we can say that "By the principle of mathematical induction, P(n) is true for all n belonging to N".

Proof:
The principle starts with a base case, which is a statement confirmed to be true for a specific starting value. Next comes the inductive step, where we assume the statement holds for some positive integer k (this is the inductive hypothesis) and then show that it must also hold for k+1. If both parts are satisfied, the statement is proven true for all required values of n.

Mathematical Induction Example
Mathematical induction is used to prove various statements. We can learn this with the help of the following example.

Example: For any positive integer n, prove that $n^3 + 2n$ is always divisible by 3

Solution:
Let P(n): $n^3 + 2n$ is divisible by 3 be the given statement.

Let P(n): $n^3 + 2n$ is divisible by 3 be the given statement.

Step 1: Basic Step

Firstly we prove that P(1) is true. Let n = 1 in $n^3 + 2n$
$= 1^3 + 2(1)$
$= 3$

As 3 is divisible by 3. Hence, P(1) is true.

Step 2: Assumption Step

Let us assume that P(k) is true
Then, $k^3 + 2k$ is divisible by 3
Thus, we can assume $k^3 + 2k = 3n$, (where n is any positive integer)....(i)

Step 3: Induction Steps

Now we have to prove that algebraic expression $(k + 1)^3 + 2(k + 1)$ is divisible by 3

$= (k + 1)^3 + 2(k + 1)$
$= k^3 + 3k^2 + 3k + 2k + 3$
$= (k^3 + 2k) + (3k^2 + 3k + 3)$

from eq(i)

$= 3n + 3(k^2 + k + 1)$
$= 3(n + k^2 + k + 1)$

As it is a multiple of 3 we can say that it is divisible by 3.

Thus, P(k+1) is true i.e. $(k + 1)^3 + 2(k + 1)$ is be divisible by 3. Now by the Principle of Mathematical Induction, we can say that, P(n): $n^3 + 2n$ is divisible by 3 is true.

Solved Examples of Mathematical Induction

Example 1: For all $n \geq 1$, prove that, $1^2 + 2^2 + 3^2 + ... + n^2 = \{n (n + 1) (2n + 1)\} / 6$

Solution:

Let the given statement be P(n),

For n = 1, the left side :
$1^2 = 1$

The right-hand side:
$$P(n) : 1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

For n=1
$$P(1) : \frac{1(1+1)(2\times1+1)}{6} = 1$$

Now, let's take a positive integer, k, and assume P(k) to be true i.e.,

$$1^2 + 2^2 + 3^2 + ... + k^2 = \frac{k(k+1)(2k+1)}{6}$$

We shall now prove that P(k + 1) is also true, so now we have,

$P(k+1) = P(k) + (k+1)^2$

$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$= (k+1)\frac{(2k^2+k)+6(k+1)}{6}$

$= \frac{(k+1)(2k^2+7k+6)}{6}$

$= \frac{(k+1)(k+2)(2k+3)}{6}$

$= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$

Thus P(k + 1) is true, whenever P(k) is true for all natural numbers. Hence, by the process of mathematical induction, the given result is true for all natural numbers.

Example 2: For all n ≥ 1, prove that, 1.2.3 + 2.3.4 + 3.4.5+...+n (n + 1) (n + 2) = {n (n + 1) (n + 2) (n + 3)} / 4

Solution:

Let the given statement be S(n),

$S(n) : 1.2.3 + 2.3.4 + 3.4.5 + \ldots + n.(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$

For n=1,

$S(1) : \frac{1(1+1)(1+2)(1+3)}{4} = 6$

which is true.

Now, let's take a positive integer, k, and assume S(k) to be true i.e.

$S(k) : 1.2.3 + 2.3.4 + 3.4.5 + \ldots + k.(k+1)(k+2) = \frac{k(k+1)(k+2)(k+3)}{4}$

We shall now prove that S(k + 1) is also true, so now we have,

$S(k+1) : S(k) + (k+1)(k+2)(k+3)$

$\Rightarrow S(k+1) : \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3)$

$\Rightarrow S(k+1) : \frac{k(k+1)(k+2)(k+3) + 4(k+1)(k+2)(k+3)}{4}$

$\Rightarrow S(k+1) : \frac{(k+1)(k+2)(k+3)(k+4)}{4}$

$\Rightarrow S(k+1) : \frac{(k+1)\{(k+1)+1\}\{(k+1)+2\}\{(k+1)+3\}}{4}$

Thus S(k + 1) is true, whenever S(k) is true for all natural numbers. And we initially showed that S(1) is true thus S(n) is true for all natural numbers.

Example 3: For all $n \geq 1$, prove that, $1 + 3 + 5 + ... + 2n - 1 = n^2$

Solution:

Let the given statement be S(n),
and $S(n) = 1 + 3 + 5 + ... + 2n - 1 = n^2$
For n = 1,
$L.H.S = 2(1) \times 1 - 1 = 1$
$R.H.S = 1^2 = 1$

Thus S(1) is true .

Now, let's take a positive integer, k, and assume S(k) to be true i.e.,
$S(k) = 1 + 3 + 5 + ... + (2k - 1) = k^2$

We shall now prove that S(k + 1) is also true, so now we have,
$1 + 3 + 5 + ... + (2(k + 1) - 1) = (k + 1)^2$

$L.H.S = 1 + 3 + 5 + ... (2k - 1) + 2k + 2 - 1$

$\Rightarrow L.H.S = S(k) + 2k + 1$
$\Rightarrow L.H.S = k^2 + 2k + 1$
$\Rightarrow L.H.S = (k + 1)^2$
$\Rightarrow L.H.S = R.H.S$

Thus S(k + 1) is true, whenever S(k) is true for all natural numbers. And we initially showed that S(1) is true thus S(n) is true for all natural numbers.

Example 4: For all $n \geq 1$, prove that, $1.2 + 2.3 + 3.4 + ... + n(n + 1) = \{n(n + 1)(n + 2)\} / 3$

Solution:

Let the given statement be S(n),

$$S(n) : 1.2 + 2.3 + 3.4 + \ldots\ldots + n.(n+1) = \frac{n(n+1)(n+2)}{3}$$

for n=1,

$$S(1) : \frac{1(1+1)(1+2)}{3} = 2$$

which is true.

Now, let's take a positive integer, k, and assume S(k) to be true i.e.,

$$S(k) : 1.2 + 2.3 + 3.4 + \ldots\ldots + k.(k+1) = \frac{k(k+1)(k+2)}{3}$$

We shall now prove that S(k + 1) is also true, so now we have,

$$S(k+1) : S(k) + (k+1)(k+2)$$

$$\Rightarrow S(k+1) : \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$\Rightarrow S(k+1) : \frac{k(k+1)(k+2)+3(k+1)(k+2)}{3}$$

$$\Rightarrow S(k+1) : \frac{(k+1)(k+2)(k+3)}{3}$$

$$\Rightarrow S(k+1) : \frac{(k+1)\{(k+1)+1\}\{(k+1)+2\}}{3}$$

Thus S(k + 1) is true, whenever S(k) is true for all natural numbers. And we initially showed that S(1) is true thus S(n) is true for all natural numbers.

Example 5: Prove that $a_n = a_1 + (n - 1)\, d$ is the general term of any arithmetic sequence.

Solution:

For n = 1,
$a_n = a_1 + (1 - 1)\, d = a_1$
so the formula holds true for n = 1

Let us assume that the formula $a_k = a_1 + (k - 1)$ is true for all natural numbers.

We shall now prove that the formula is also true for k+1, so now we have,
$a_{k+1} = a_1 + [(k + 1) - 1]\, d = a_1 + k \cdot d.$

We assumed that $a_k = a_1 + (k - 1)\, d$, and by the definition of an arithmetic sequence $a_{k+1} - a_k = d$,

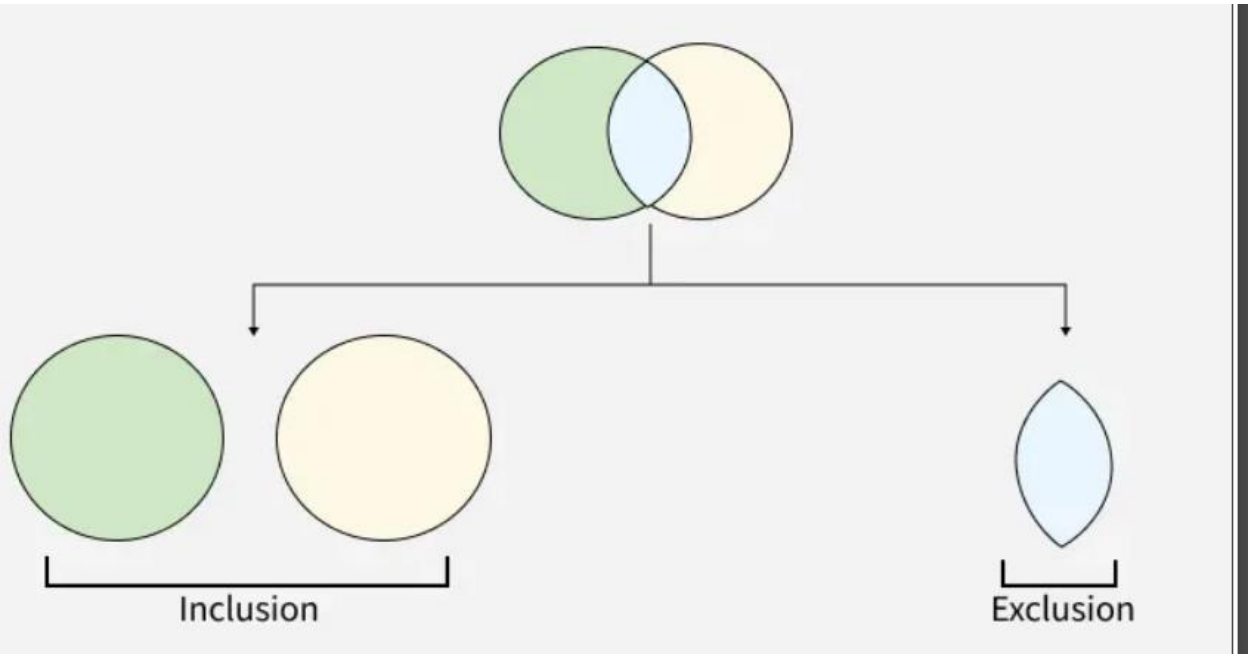
Then, $a_{k+1} - a_k$
$= (a_1 + k \cdot d) - (a1 + (k - 1)d)$
$= a_1 - a_1 + kd - kd + d$
$= d$

Thus, the formula is true for k + 1, whenever it is true for k. And we initially showed that the formula is true for n = 1. Thus, the formula is true for all natural numbers.

Principle of Inclusion and Exclusion

The **Principle of Inclusion and Exclusion (PIE)** is a counting technique used to find the size of the union of multiple sets by systematically including and excluding overlapping parts.

It corrects the overcounting that happens when you simply add the sizes of the sets. When sets have overlapping elements, counting them directly can lead to overcounting. The Principle of Inclusion and Exclusion fixes this by alternately adding and subtracting the sizes of the intersection to get the exact total.



For example, when calculating the number of people who own a dog or a cat, PIE adds up all dog owners and subtracts all cat owners. This includes:

- Adding the sizes of each individual set.
- Subtracting the sizes of the pairwise intersections to avoid double-counting.

- Adding back the size of the intersection of all three sets, as it was subtracted too many times.
- Subtracting the sizes of the intersection of four sets, as it was added twice in the last step.
- And so on.

Formula for Principle of Inclusion & Exclusion

In general, for n sets $A_1, A_2, \ldots, A_n$:

$$\left|A_1 \cup A_2 \cup \cdots \cup A_n\right| = \sum_{i=1}^{n} \left|A_i\right| - \sum_{1 \le i < j \le n} \left|A_i \cap A_j\right| + \sum_{1 \le i < j < k \le n} \left|A_i \cap A_j \cap A_k\right| - \cdots + (-1)^{n+1} \left|A_1 \cap A_2 \cap \cdots \cap A_n\right|$$

Where:

- The first term sums the sizes of all individual sets.
- The second term subtracts the sizes of all pairwise intersections.
- The third term adds back the sizes of all three-way intersections.
- This alternating pattern continues, with the signs changing for each subsequent term, until the intersection of all N sets is reached.

Note: This formula effectively accounts for all overlaps among the sets, ensuring that each unique element is counted exactly once.

Formula for Two Sets

For two sets A and B

|AUB| = |A|+|B|-|A∩B|

Where,

- |A| is the number of elements in set *A*.
- **|**B| is the number of elements in set B.
- |A∩B| is the number of elements in both set A and set B.

Formula for Three Sets

For three sets A,B, and C, the Principle of Inclusion and Exclusion (PIE) formula to find the size of the union is:

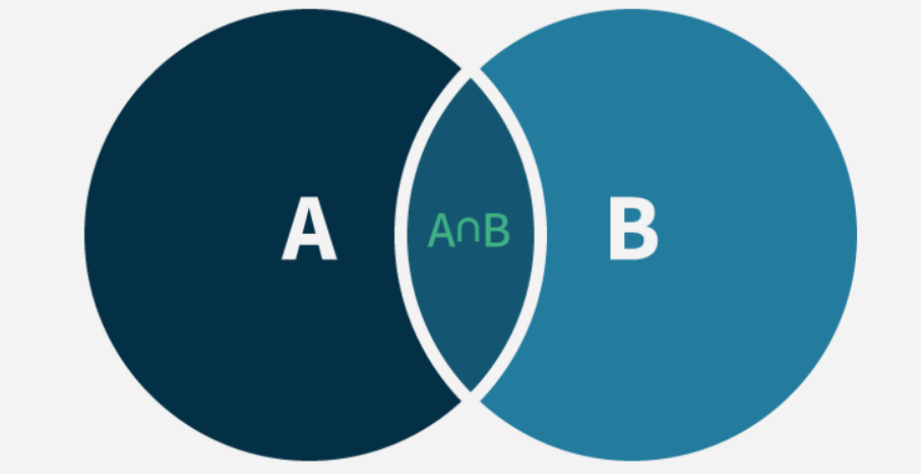
|AUBUC| = |A|+|B|+|C|-|A∩B|-|A∩C|-|B∩C|+|A∩B∩C|

Where,

- |A|, |B|, and |C| are the number of elements in the sets A, B, and C, respectively.
- |A∩B|, |A∩C|, |and B∩C| are the numbers of elements in the pairwise intersections of the sets.
- |A∩B∩C| is the number of elements that are in all three sets.

Derivation of the Inclusion and Exclusion Principle Using Venn Diagrams
We can visualize this using a Venn diagram for a finite number of sets. Here we will discuss the formula for the Principle of Inclusion and Exclusion for two and three sets and derive it using a Venn diagram.

Venn Diagram for Two Sets

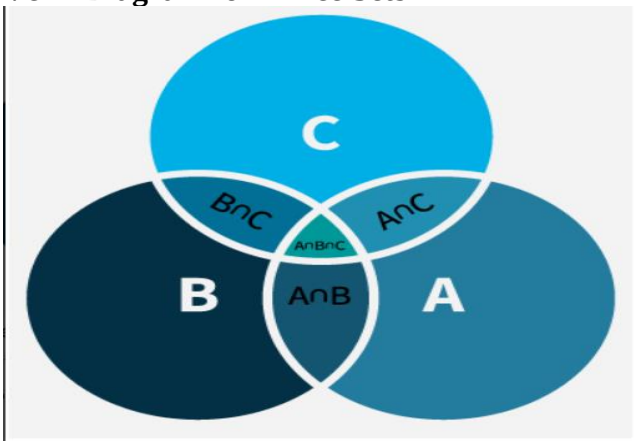


- **First Step (Inclusion):** Add the sizes of the two sets |A| and |B|. This counts all the elements in A and B, but elements in the intersection of A∩and B are counted twice (once in A and once in B).

- **Second Step (Exclusion):** Subtract the size of the intersection |A∩B||A \cap B||A∩B|, because these elements were counted twice in the first step.

Thus, the number of elements in the union of these two sets, |A ∪ and B|, is given by:
|A∪B| = |A| + |B| − |A∩B|

Venn Diagram for Three Sets



- **First Step (Inclusion):** Add the sizes of the individual sets |A|, |B|, and |C|. This counts all the elements in A, B, and C, but it also counts the elements in the intersections more than once.

- **Second Step (Exclusion):** Subtract the pairwise intersections |A∩B|, |B∩C|, and |C∩A| because these elements were counted twice in the first step.

- **Third Step (Re) Inclusion):** Add back the intersection of all three sets |A∩, B∩C|, as it was subtracted three times (once for each pairwise intersection) in the previous step.

Thus, the number of elements in the union of these three sets, |A∪B∪C|, is given by:
|A∪B∪C| = |A| + |B| + |C| − |A∩B| − |B∩C| − |C∩A| + |A∩B∩C|

Solved Examples
Question 1: In a class of 100 students:
- 60 study Math.
- 45 study Science.
- 20 students study both Math and Science.

How many students study either Math or Science?

Solution:
As we know, n(A∪B) = n(A) + n(B) – n(A∩B)
Given:
- n(A) = 60
- n(B) = 45
- n(A∩B) = 20

n(A∪B) = n(A) + n(B) – n(A∩B) = 60 + 45 - 20 = 85
Thus, 85 students study either Math or Science.

Question 2: The probability of getting at least one head is Problem: In a survey of 120 people:
- 80 people like tea.
- 70 people like coffee.
- 50 people like both tea and coffee.

How many people like either tea or coffee?

Solution:
As we know, n(A∪B) = n(A) + n(B) – n(A∩B)
Given:
- IAI = 80
- IBI = 70
- IA∩BI = 50

Thus, n(A∪B) = 80 + 70 - 50 = 100
Thus, 100 people like either tea or coffee.

Question 3: A company has 30 employees, 10 are assigned to Task A, 15 to Task B, and 5 are assigned to both. How many employees are assigned to at least one task?

Solution:
As we know, n(A∪B) = n(A) + n(B) – n(A∩B)

Given:

- IAI = 10 Task A
- IBI = 15 Task B
- IA∩BI = 5 Both tasks

Substitute the Values, We get
n(A∪B) = 10 + 15 - 5 = 20
Thus, 20 employees are assigned to at least one task.

Question 4: In a survey of 200 people:

- 120 likes pizza.
- 100 like burgers.
- 80 like tacos.
- 60 likes both pizza and burgers.
- 40 likes both burgers and tacos.
- 30 likes both pizza and tacos.
- 20 likes all three: pizza, burgers, and tacos.

How many people like at least one of these three foods?

Solution:
As we know, |AUBUC| = |A| + |B| + |C| - |A∩B| - |A∩C| - |B∩C| + |A∩B∩C|
Given:

- |A| = 120 Pizza lovers
- |B| = 100 Burger lovers
- |C| = 80 Taco lovers
- |A∩B| = 60
- |A∩C| = 30
- |B∩C| = 40
- |A∩B∩C| = 20

Substituting in the formula, we get,
|AUBUC| = 120 + 100 + 80 - 60 - 30 - 40 + 20 = 190
Thus, 190 people like at least one of the three foods.

Question 5: In a university of 300 students:

- 150 students take Math.
- 120 students take Physics.
- 100 students take Chemistry.
- 80 students take both Math and Physics.
- 60 students take both Physics and Chemistry.
- 50 students take both Math and Chemistry.
- 30 students take all three subjects.

How many students are taking at least one subject?

Solution:
As we know, |AUBUC| = |A| + |B| + |C| - |A∩B| - |A∩C| - |B∩C| + |A∩B∩C|

Given:

- |A| = 150 (Math),
- |B| = 120 (Physics),
- |C| = 100 (Chemistry),
- |A∩B| = 80
- |A∩C| = 60
- |B∩C| = 50
- |A∩B∩C| = 30

Substituting values, we get
|AUBUC| = 150 + 120 + 100 - 80 - 60 - 50 + 30 = 210
Thus, 210 students are taking at least one subject.

Permutations and Combinations
In combinatorics, we study methods of **counting arrangements and selections** of objects. Two basic tools are:

- **Permutation:** arrangement of objects in a specific order.
- **Combination:** selection of objects without regard to order.

Fundamental Counting Principle
If an event can occur in mmm ways, and another independent event in nnn ways, then both events can occur in m×n ways.

Example:

- 3 shirts and 2 trousers → total outfits = 3×2=6

Definition
A permutation is an ordered arrangement of objects.

- Order matters.
- Formula for permutations of n objects taken r at a time:
- P (n, r) = $\frac{n!}{(n-r)!}$

Permutations without Repetition

- Each object is used once.
- Example: How many ways can we arrange 3 letters A, B, C?

  P (3, 3) = $\frac{3!}{(3-3)!}$=6

  Arrangements: ABC, ACB, BAC, BCA, CAB, CBA

Worked Examples
Example 1 — Permutation without repetition
How many 4-digit PINs can be formed from digits 0–9 if no digit is repeated?

P (10, 4) $= \frac{10!}{(10-4)!} = \frac{10!}{6!} = 5040$

Permutations with Repetition

- Objects may be repeated.
- Formula: n^r.

How many ways can we arrange 3 letters A, B, C using permutation *with* repetition?

Step 1: Understand the problem

- We have **3 letters**: A, B, C
- We want to form **arrangements of length 3**.
- Since **repetition is allowed**, each position can be filled with any of the 3 letters.

Step 2: Apply formula for permutation with repetition

Total arrangements=n^r

Where,

- n= number of objects = 3
- r= length of arrangement = 3

  $3^3=27$

Step 3: Example listing

Some examples:

- AAA, AAB, AAC
- ABA, ABB, ABC
- BAA, BBB, BCC
- CCC, CCA, CCB …and so on, total = **27 distinct arrangements**.

Example 2 — Permutation with repetition

How many 4-digit PINs can be formed from digits 0–9 if digits may repeat?

$10^4=10,000$

Combinations

A **combination** is a selection of objects **without regard to order**.

- Formula:

  C (n, r) $= \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Combinations without Repetition

- Choose r distinct objects from n.
- Example: Number of ways to choose 3 students from 10:

  $\binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10!}{3!.7!} = 120$

Combinations with Repetition

- Objects can be chosen more than once.
- Formula:

$$\binom{n+r-1}{r}$$

where n= number of object types, r = number of choices.

- Example: Number of ways to choose 3 fruits from {apple, banana, orange} (repetition allowed):

$$\binom{3+3-1}{3}=\binom{5}{3}=10$$

Example 3 — Combination without repetition

From a group of 12 students, how many ways can we select a committee of 4?

$$\binom{12}{4}=\frac{12!}{4!(12-4)!}=\frac{12!}{4!.8!}=495$$

Example 4 — Combination with repetition

How many ways can we distribute 5 identical candies to 3 children?

$$\binom{3+5-1}{5}=\binom{7}{5}=21$$

Example 5 — Mixed problem

How many distinct arrangements of the word "MISSISSIPPI"? Total letters = 11 (M=1, I=4, S=4, P=2).

$$\frac{11!}{1!.4!.\ 4!.\ 2!}=\frac{39916800}{1152}=34{,}650$$

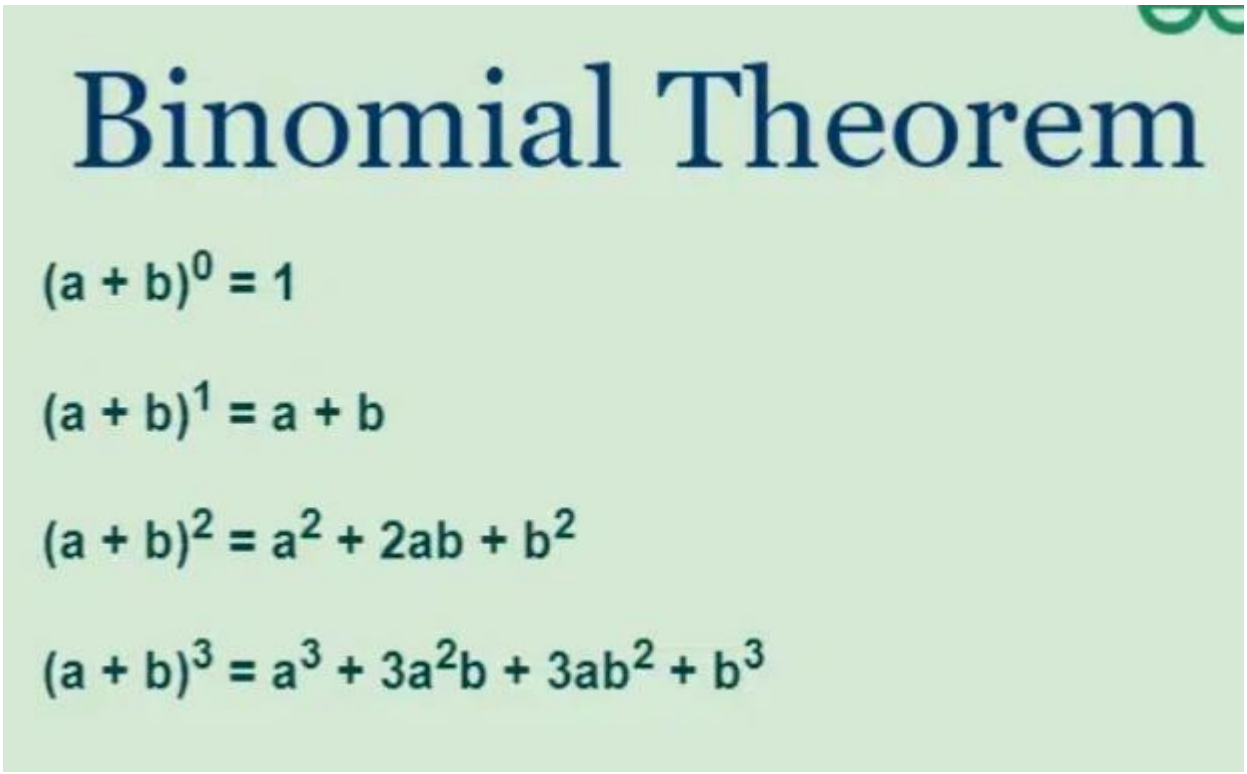
Example 6:

Number of distinct arrangements of word "BALLOON" (7 letters: B=1, A=1, L=2, O=2, N=1):

$$\frac{7!}{1!.1!.\ 2!.\ 2!.\ 1!}=\frac{5040}{4}=1{,}260$$

Binomial Theorem

Binomial theorem is a fundamental principle in algebra that describes the algebraic expansion of powers of a binomial. According to this theorem, the expression $(a + b)^n$ where a and b are any numbers and n is a non-negative integer. It can be expanded into the sum of terms involving powers of a and b. Binomial theorem is used to find the expansion of **two terms** hence it is called the Binomial Theorem.



Binomial Theorem Statement

Binomial theorem for the expansion of $(a+b)^n$ is stated as,

$$\mathbf{(a + b)^n = {}^nC_0\, a^n b^0 + {}^nC_1\, a^{n-1} b^1 + {}^nC_2\, a^{n-2} b^2 + ... + {}^nC_r\, a^{n-r} b^r + ... + {}^nC_n\, a^0 b^n}$$

where $n > 0$ and the ${}^nC_k$ is the binomial coefficient.

Example: Find the expansion of $(x + 5)^6$ using the binomial theorem.

Solution:

We know that here,

$(a + b)^n = {}^nC_0\, a^n b^0 + {}^nC_1\, a^{n-1} b^1 + {}^nC_2\, a^{n-2} b^2 + ... + {}^nC_r\, a^{n-r} b^r + ... + {}^nC_n\, a^0 b^n$

Thus, $(x + 5)^6 = {}^6C_0\, x^6 5^0 + {}^6C_1\, x^{6-1} 5^1 + {}^6C_2\, x^{6-2} 5^2 + {}^6C_3\, x^{6-3} 5^3 + {}^6C_4\, x^{6-4} 5^4 + {}^6C_5\, x^{6-5} 5^5 + {}^6C_6\, x^{6-6} 5^6$

$= {}^6C_0\, x^6 + {}^6C_1\, x^5 5 + {}^6C_2\, x^4 5^2 + {}^6C_3\, x^3 5^3 + {}^6C_4\, x^2 5^4 + {}^6C_5\, x^1 5^5 + {}^6C_6\, x^0 5^6$

$= x^6 + 30x^5 + 375x^4 + 2500x^3 + 9375x^2 + 18750x + 15625$

Binomial Expansion Formula

Binomial theorem formula gives the expansion of the algebraic identities in the form of a series. This formula is used to find the expansion up to n terms of the $(a+b)^n$. The binomial expansion of $(a + b)^n$ can easily be represented with the summation formula.

Binomial Theorem Formula for the expansion of $(a + b)^n$ is, $\mathbf{(a + b)^n = \sum_r^n {}^nC_r\, a^{n-r} b^r}$

where,

- **n** is a positive integer,
- **a, b** are real numbers, and $\mathbf{0 < r \leq n}$

We can easily find the expansion of the various identities such as $(x+y)^7$, $(x+9)^{11}$, and others using the Binomial Theorem Formula. We can also find the expansion of $(ax + by)^n$ using the Binomial Theorem Formula, Expansion formula for $(ax + by)^n$ is,

$$\mathbf{(ax + by)^n = \sum^{n}_{r} {}^nC_r\ (ax)^{n-r}\ (by)^r}$$

where $\mathbf{0< r \leq n.}$

Binomial Theorem Proof

We can easily proof Binomial Theorem Expansion using the concept of Principle of Mathematical Induction

Let's take x, a, n ∈ N, and then the Binomial Theorem says that,

$$\mathbf{(x+y)^n = {}^nC_0\ x^ny^0 + {}^nC_1\ x^{n-1}y^1 + {}^nC_2\ x^{n-2}\ y^2 + ... + {}^nC_{n-1}\ x^1y^{n-1} + {}^nC_n\ x^0y^n}$$

Proof:

For n = 1, we get

$(x +y)^1 = x +y$ which is true.

For n = 2,

$(x +y)^2 = (x + y)\ (x +y)$

$\Rightarrow (x +y)^2 = x^2 + xy + xy + y^2$ (using distributive property of multiplication over addition)

$\Rightarrow (x +y)^2 = x^2 + 2xy + y^2$

which is also true.

Thus, theorem is true for n = 1 and n = 2.

Let's take a positive integer k.

$(x+y)^k = {}^kC_0\ x^ky^0 + {}^kC_1\ x^{k-1}y^1 + {}^kC_2\ x^{k-2}\ y^2 + ... + {}^kC_{k-1}\ x^1y^{k-1} + {}^kC_k\ x^0y^k$

Now consider the expansion for n = k + 1

$(x + y)^{k+1} = (x + y)\ (x + y)^k$

$\Rightarrow (x + y)^{k+1} = (x + y)\ (x^k + {}^kC_1\ x^{k-1}y^1 + {}^kC_2\ x^{k-2}\ y^2 + ... + {}^kC_r\ x^{k-r}y^r + ... + y^k)$

$\Rightarrow (x + y)^{k+1} = x^{k+1} + (1 + {}^kC_1)x^ky + ({}^kC_1 + {}^kC_2)\ x^{k-1}y^2 + ... + ({}^kC_{r-1} + {}^kC_r)\ x^{k-r+1}y^r + ... + ({}^kC_{k-1} + 1)\ xy^k + yk^{+1}$

$\Rightarrow (x + y)^{k+1} = x^{k+1} + {}^{k+1}C_1x^ky + {}^{k+1}C_2\ x^{k-1}y^2 + ... + {}^{k+1}C_r\ x^{k-r+1}y^r + ... + {}^{k+1}C_k\ xy^k + y^{k+1}$

[As we know, ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$]

This result is true for n = k+1.

Binomial Expansion

Binomial Theorem is used to expand the algebraic identity $(x + y)^{n.}$ Hence it is also called the binomial expansion. The binomial expansion of $(x + y)^n$ with the help of the binomial theorem is, (x+y) n = nC0 xny0 + nC1 xn-1y1 + nC2 xn-2 y2 + ... + nCn-1 x1yn-1 + nCnx0yn

Using this expansion we get each term in the expansion of $(x+y)^{nl}$

Example: Find the value of $(x+y)^2$ and $(x+y)^3$ using Binomial expansion.

Solution:

$(x+y)^2 = {}^2C_0\ x^2y^0 + {}^2C_1\ x^{2-1}y^1 + {}^2C_2\ x^{2-2}\ y^2$

$\Rightarrow (x+y)^2 = x^2 + 2xy + y^2$

And $(x+y)^3 = {}^3C_0\ x^3y^0 + {}^3C_1\ x^{3-1}y^1 + {}^3C_2\ x^{3-2}\ y^2 + {}^3C_3\ x^{3-3}\ y^3$

$\Rightarrow (x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$
$\Rightarrow (x+y)^3 = x^3 + 3xy(x+y) + y^3$

Discrete Probability Distribution

A discrete probability distribution describes the likelihood of each possible outcome for a discrete random variable. A discrete random variable is a variable that can take on a countable number of distinct values, typically whole numbers. In simpler terms, it's a way to assign a probability to each value that the variable can have, showing how likely each outcome is. The common examples of discrete probability distributions include Bernoulli, Binomial, Poisson, and Geometric distributions.

Basic Concepts

Bernoulli Distribution

A discrete probability distribution with the probability of success p if the value of random variable is 1 and the probability of failure 1-p if the value of random variable is zero is called the Bernoulli distribution. The probability mass function of the Bernoulli distribution is given by:

$$P(X = x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$$

Binomial Distribution

A discrete probability distribution that includes the number of trials n, probability of success and probability of failure is called as Binomial distribution. The probability mass function of the Binomial distribution is given by:

$$P(X = x) = {}^{n}C_{x}\, p^{x}\, (1-p)^{n-x}$$

Poisson Distribution

A discrete probability distribution that gives the number of events occurred at a specific time period with the help of its mean is called as the Poisson distribution. The probability mass function of Poisson distribution is given by:

$$P(X = x) = [\lambda^{x} \times e^{-\lambda}] / x!$$

Geometric Distribution

A discrete probability distribution that includes the successive failure probability until the success probability is encountered is called as Geometric distribution. The probability mass function of the geometric distribution is given by:

$$P(X = x) = (1 - p)^{x}p$$

Random Experiment: An experiment in which all possible outcomes are known and the exact output cannot be predicted in advance is called a random experiment. Tossing a fair coin is an example of random experiment.

Sample Space: When we perform an experiment, then the set S of all possible outcomes is called the sample space. If we toss a coin, the sample space $S = \{H, T\}$

Event: Any subset of a sample space is called an event. After tossing a coin, getting Head on the top is an event.

The word "probability" means the chance of occurrence of a particular event. The best we can say is how likely they are to happen, using the idea of probability.

Probability of occurrence of an event $= \frac{Total\ number\ of\ favourable\ outcome}{Total\ number\ of\ Outcomes}$

As the occurrence of any event varies between 0% and 100%, the probability varies between 0 and 1.

Throwing a Dice
When a dice is thrown, six possible outcomes can be on the top − 1,2,3,4,5,6
The probability of any one of the numbers is 1/6
The probability of getting even numbers is 3/6 = 1/2
The probability of getting odd numbers is 3/6 = 1/2

Taking Cards from a Deck
From a deck of 52 cards, if one card is picked find the probability of an ace being drawn and also find the probability of a diamond being drawn.
Total number of possible outcomes − 52
Outcomes of being an ace − 4
Probability of being an ace = 4/52 = 1/13
Probability of being a diamond = 13/52 = 1/4

Probability Axioms

- The probability of an event always varies from 0 to 1. $[0 \leq P(x) \leq 1]$
- For an impossible event the probability is 0 and for a certain event the probability is 1.
- If the occurrence of one event is not influenced by another event, they are called mutually exclusive or disjoint.
  If $A_1, A_2 \ldots . A_n$ are mutually exclusive/disjoint events, then $P(A_i \cap A_j) = \emptyset$ for $i \neq j$ and $P(A_1 \cup A_2 \cup \ldots . A_n) = P(A_1) + P(A_2) + \ldots . P(A_n)$

Properties of Probability

- If there are two events x and $\overline{x}$which are complementary, then the probability of the complementary event is –

$$p(\overline{x}) = 1 - p(x)$$

- For two non-disjoint events A and B, the probability of the union of two events – $P(A \cup B) = P(A) + P(B)$
- If an event A is a subset of another event B (i.e. $A \subset B$), then the probability of A is less than or equal to the probability of B. Hence, $A \subset B$ implies $P(A) \leq p(B)$

Conditional Probability

The conditional probability of an event B is the probability that the event will occur given an event A has already occurred. This is written as $P(B|A)$

.

Mathematically – $P(B|A) = P(A \cap B)/P(A)$

If event A and B are mutually exclusive, then the conditional probability of event B after the event A will be the probability of event B that is $P(B)$.

OR

Conditional Probability

The probability of event A given that event B has occurred:

P (A\B) = $\frac{P\ (A \cap B)}{P(B)}, P\ (B)\ >\ 0$

Worked Examples

Example 1 — Simple probability

A die is rolled. Find probability of getting a number greater than 4.

- Sample space S= {1,2,3,4,5,6}.
- Favorable outcomes = {5,6} → 2

- Probability: P (E)= $\frac{2}{6} = \frac{1}{3}$

Example 2 — Conditional probability
A box contains 3 red and 2 blue balls. One is chosen at random. Find probability it is red, given that it is not blue.

- P (Red) = 3/5
- P (Not blue) = P (Red) = 3/5
- So, P (Red\Not blue) = (3/5) / (3/5) = 1

Problem 1
In a country 50% of all teenagers own a cycle and 30% of all teenagers own a bike and cycle. What is the probability that a teenager owns bike given that the teenager owns a cycle?

Solution
Let us assume A is the event of teenagers owning only a cycle and B is the event of teenagers owning only a bike.
So, $P(A)=50/100=0.5$
and $P(A\cap B) =30/100=0.3$
from the given problem.
$P(B|A) =P(A\cap B)/P(A)=0.3/0.5=0.6$
Hence, the probability that a teenager owns bike given that the teenager owns a cycle is 60%.

Problem 2
In a class, 50% of all students play cricket and 25% of all students play cricket and volleyball. What is the probability that a student plays volleyball given that the student plays cricket?

Solution
Let us assume A is the event of students playing only cricket and B is the event of students playing only volleyball.
So, $P(A)=50/100=0.5$
and $P(A\cap B) =25/100=0.25$
from the given problem.
$P[B|A]=P[A\cap B]/P[A]=0.25/0.5=0.5$
Hence, the probability that a student plays volleyball given that the student plays cricket is 50%.

Problem 3
Six good laptops and three defective laptops are mixed up. To find the defective laptops all of them are tested one-by-one at random. What is the probability to find both of the defective laptops in the first two pick?

Solution
Let A be the event that we find a defective laptop in the first test and B be the event that we find a defective laptop in the second test.
Hence, $P(A\cap B) =P(A)P(B|A) =3/9\times2/8=1/12$

Bayes' Theorem

Theorem − If A and B are two mutually exclusive events, where *P*(*A*) is the probability of A and *P*(*B*) is the probability of B, *P*(*A*|*B*) is the probability of A given that B is true. *P*(*B*|*A*) is the probability of B given that A is true, then Bayes Theorem states:

$$P(A\backslash B) = \frac{P(B\backslash A)P(A)}{\sum_{i=1}^{n} P(B\backslash Ai)P(Ai)}$$

Problem

Consider three pen-stands. The first pen-stand contains 2 red pens and 3 blue pens; the second one has 3 red pens and 2 blue pens; and the third one has 4 red pens and 1 blue pen. There is equal probability of each pen-stand to be selected. If one pen is drawn at random, what is the probability that it is a red pen?

Solution

Let *Ai* be the event that i^{th} pen-stand is selected.
Here, i = 1,2,3.
Since probability for choosing a pen-stand is equal, *P*(*Ai*)=1/3
Let B be the event that a red pen is drawn.
The probability that a red pen is chosen among the five pens of the first pen-stand,
P(*B*|*A*1) =2/5
The probability that a red pen is chosen among the five pens of the second pen-stand,
P(*B*|*A*2) =3/5
The probability that a red pen is chosen among the five pens of the third pen-stand,
P(*B*|*A*3) =4/5
According to Bayes' Theorem,
P(*B*)=*P*(*A*1). *P*(*B*|*A*1) +*P*(*A*2). *P*(*B*|*A*2) +*P*(*A*3). *P*(*B*|*A*3)
=1/3.2/5+1/3.3/5+1/3.4/5
=3/5